



Data harmonization for
neuroscientific research:
Theory, challenges, and
applications.

*Emma Prevot, Jacob Turnbull
Gaurav Bhalerao , Johanna Bayer ,
Nicolás Nieto.*

OHBM 2026 DISCLOSURES

None of the speakers have any conflicts of interest in relation to this presentation to disclose.



Estimated Schedule

- Block 1 - Introduction to Effects of Site
- Block 2 - Fundamentals of data harmonization
- Block 3 - Location-Scale (LS) models. (First Half)
- Coffee break
- Block 3 - Alternatives to ComBat (Second half)
- Block 4 - Future Direction



About the speakers



Emma Prevot
PhD student
University of Oxford



Jake Turnbull
PhD student
University of Oxford



Nico Nieto
Postdoctoral Researcher
Research Center Jülich



Johanna Bayer
Postdoctoral Fellow
McGill University

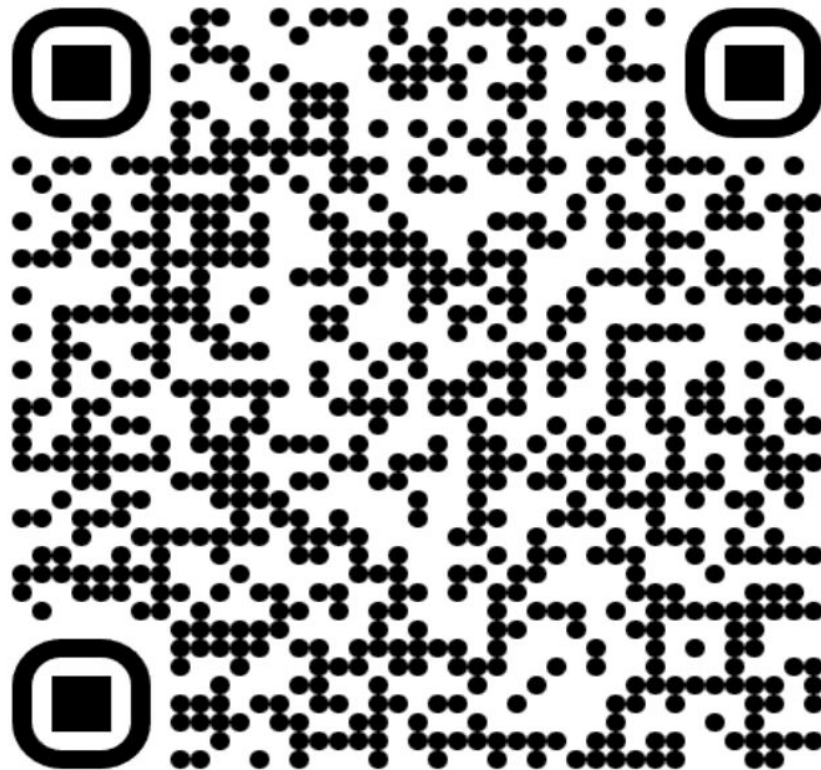


Gaurav Bhalerao
Postdoctoral Researcher
University of Oxford



Getting started!

GitHub Repository



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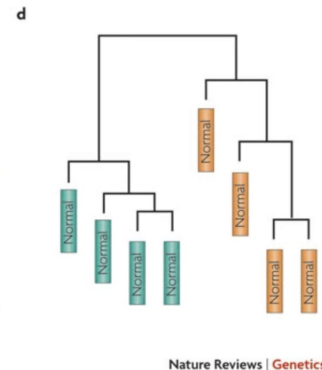
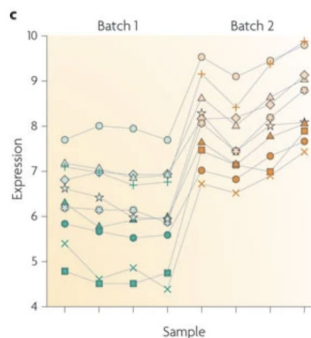
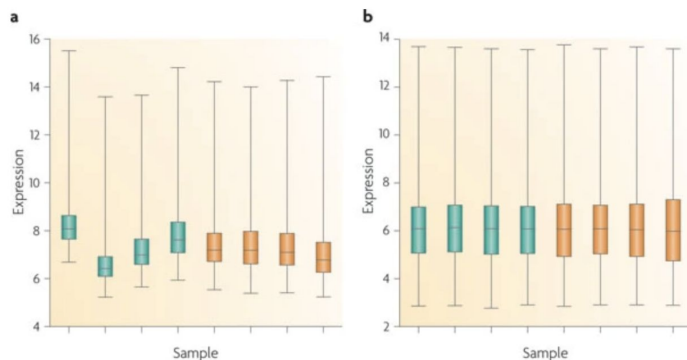
BLOCK 1



What is a site effect?

- **Batch effects** - originally known from genomics (Leek 2010)
 - Analysis done on a different day, time point or by different lab technician

Figure 1: Demonstration of normalization and surviving batch effects.



What is a site effect?

- **Batch effects** - originally known from genomics (Johnson, 2007; Leek 2010)
 - Analysis done on a different day, time point or by different lab technician

Adjusting batch effects in microarray expression data using empirical Bayes methods

W. EVAN JOHNSON, CHENG LI*

*Department of Biostatistics and Computational Biology,
Dana-Farber Cancer Institute, Boston, MA, USA and Department of Biostatistics,
Harvard School of Public Health, Boston, MA, USA
cli@hsph.harvard.edu*



What is a site effect?

- **Batch effects** - originally known from genomics (Fortin 2017)
 - Analysis done on a different day, time point or by different lab technician

Harmonization of multi-site diffusion tensor imaging data

Jean-Philippe Fortin^{1,*}, Drew Parker², Birkan Tunç², Takanori Watanabe², Mark A. Elliott³, Kosha Ruparel⁴, David R. Roalf⁴, Theodore D. Satterthwaite⁴, Ruben C. Gur^{3,4}, Raquel E. Gur^{3,4}, Robert T. Schultz⁵, Ragini Verma^{2,†}, and Russell T. Shinohara^{1,†}

¹Department of Biostatistics and Epidemiology, Perelman School of Medicine, University of Pennsylvania

²Center for Biomedical Image Computing and Analytics, Department of Radiology, Perelman School of Medicine, University of Pennsylvania

³Department of Radiology, Perelman School of Medicine, University of Pennsylvania

⁴Department of Psychiatry, Perelman School of Medicine, University of Pennsylvania

⁵Center for Autism Research, The Children's Hospital of Philadelphia



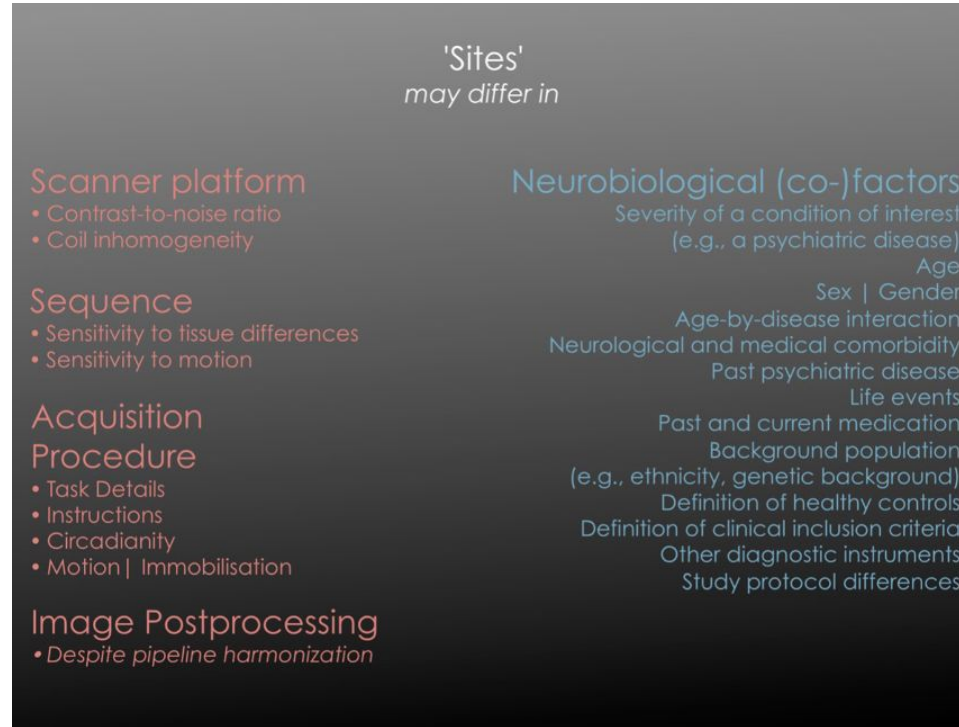
What causes a “site effect”?

Bayer, J. M. M., (2022). Site effects how-to and when: An overview of retrospective techniques to accommodate site effects in multi-site neuroimaging analyses. *Frontiers in Neurology*, 13, 923988.



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What causes a “site effect”?



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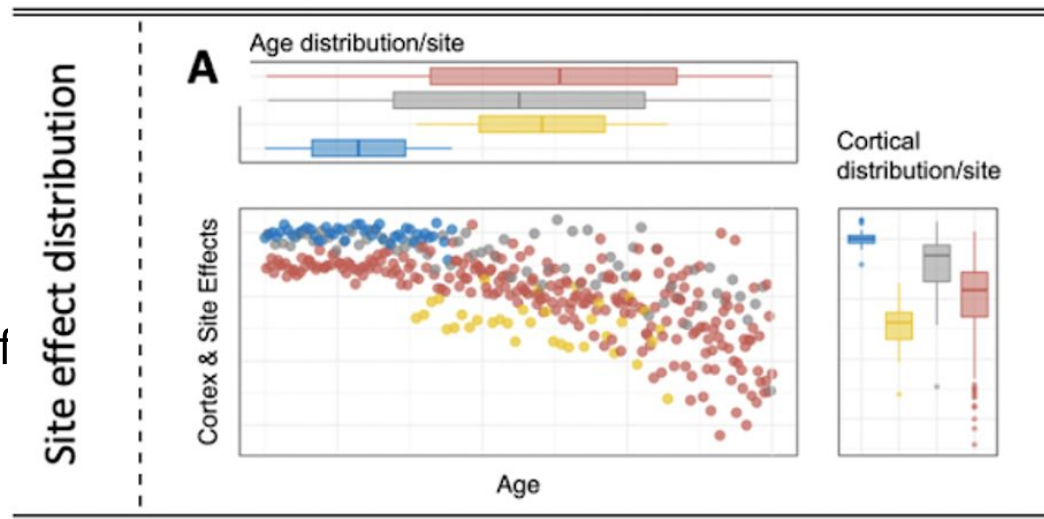
Why are site effects a problem?

- Tension between creating large data sets from pooling and site effects
- Prevent generalisability of results across studies
- Often overshadow (small) clinical effects
- Can interact with effects of interest and artificially increase potential effects of interest
- Open data sets often contain multiple studies



How “site” can interact with your data

- Mean shift (intercept offset)
- Difference in variance
- Interaction between site and biological association (different slopes for different sites)
- Interaction between variance of biological variable and site
- Linear and non-linear effects



BLOCK 2



What is site (data) harmonization

- Effects of site in your data
 - That remain after normalisation (Leek 2010)
- Adjustment step that makes data collected at different sites, scanners, batches, or platforms behave as if they came from a common measurement process, while trying to preserve the biological variation
- Neuroimaging, the problem is scanner/vendor/protocol/site differences
- Genomics, it is batch, platform, and processing variation.



Motivation for site harmonization

- Reproducible and robust results!
- Data pooling
 - Increasing the statistical power by being able to pool data across sites
 - Using open data sets
- Generalizability
 - Being able to estimate effects of interest safely across populations and data sets



Motivation for site harmonization

- Ideally: biological variance remains/ effects increase
- Site variance goes down/cannot be detected
- More robust results
- Generalizable results



Motivation for LEARNING about site harmonization

- Getting an overview of different tools and their benefits and disadvantages
- Understanding which tool should be used for which data type
- Understanding how site can interact with other variables of interest
- Understanding your data (and complex effects in it) better



Video on Harmonization evaluation and test-retest dataset



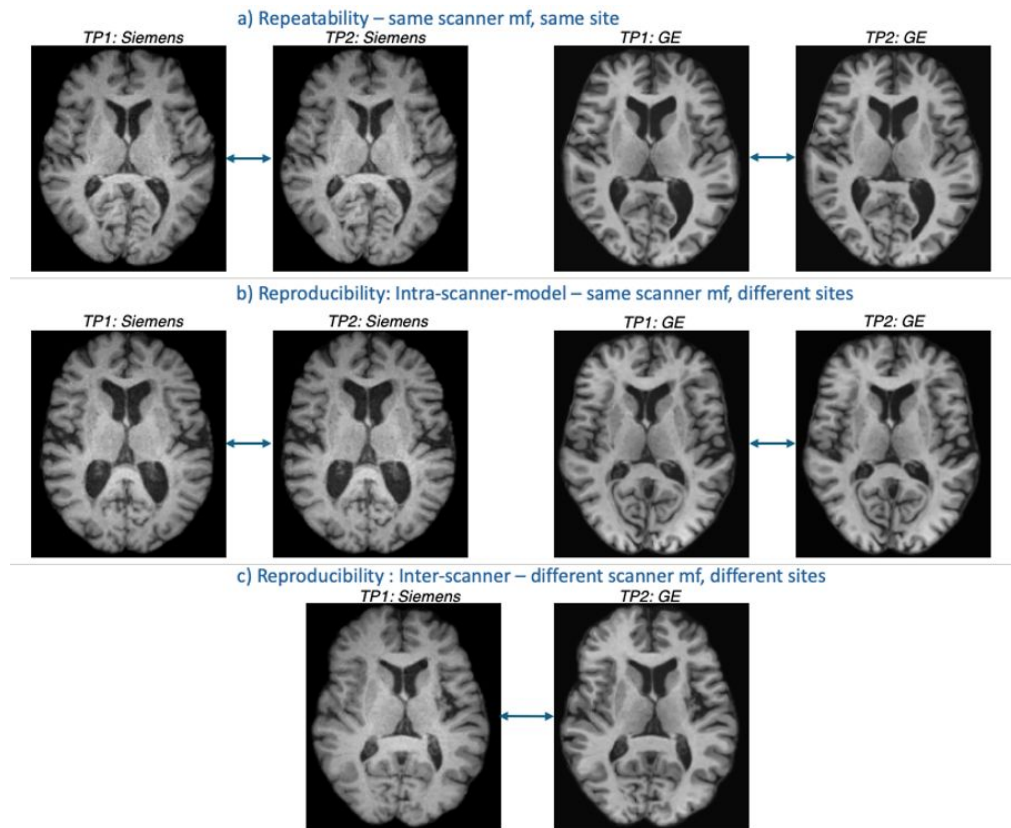
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Harmonisation Goals Depend on Study Design

Parameter	Cross-sectional	Longitudinal (expected change)	Test–retest (no expected change)
What is expected	Measurements may differ between individuals/groups	Measurements should change meaningfully (e.g., atrophy)	Measurements should be stable
Role of batch effect	Create artificial group differences or mask true group effects	Distort or obscure true change	Introduce artificial differences
Consequence	Biased group comparisons and reduced generalisability	Masked or spurious longitudinal effects	Reduced measurement stability
What we want	Accurate comparison between groups/sites	Accurate detection of true change	High consistency across scans
Example analysis	Dementia vs control classification; association with diagnosis, cognition or genetics	Rate of hippocampal atrophy; cognitive decline; disease progression modelling	Scan–rescan reliability; measurement reproducibility; biomarker stability



Scanner/Site variability using scan-rescan example

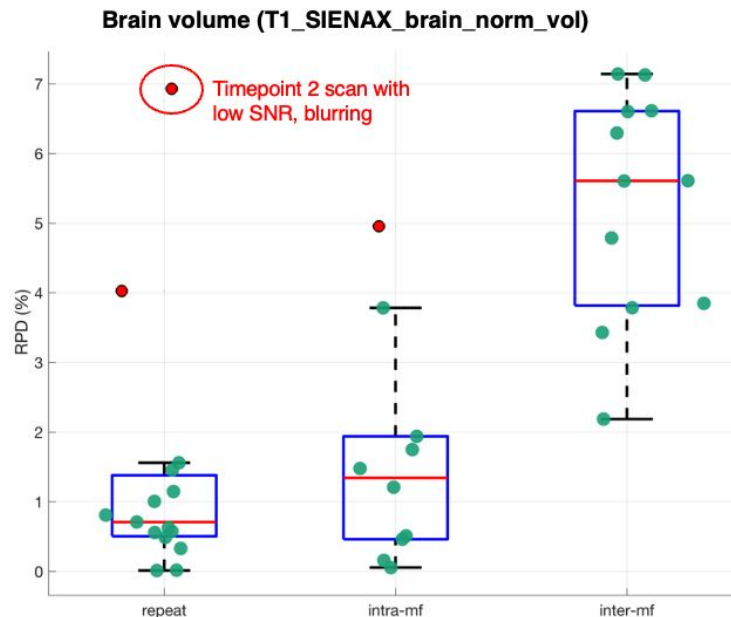
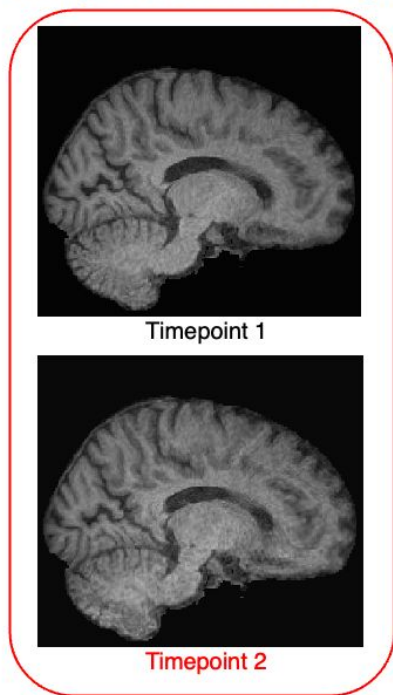


- 37 subjects
- 2 timepoints
- 2 scanners - Siemens, GE
- 8 sites
- 3 groups - repeat (15), intra (10) and inter-scanner (12)

Dementias Platform UK PET-MR dataset -
<https://doi.org/10.64898/2026.04.21.26351106>

How does it impact subject variability?

Quality of the scans have high impact on the variability



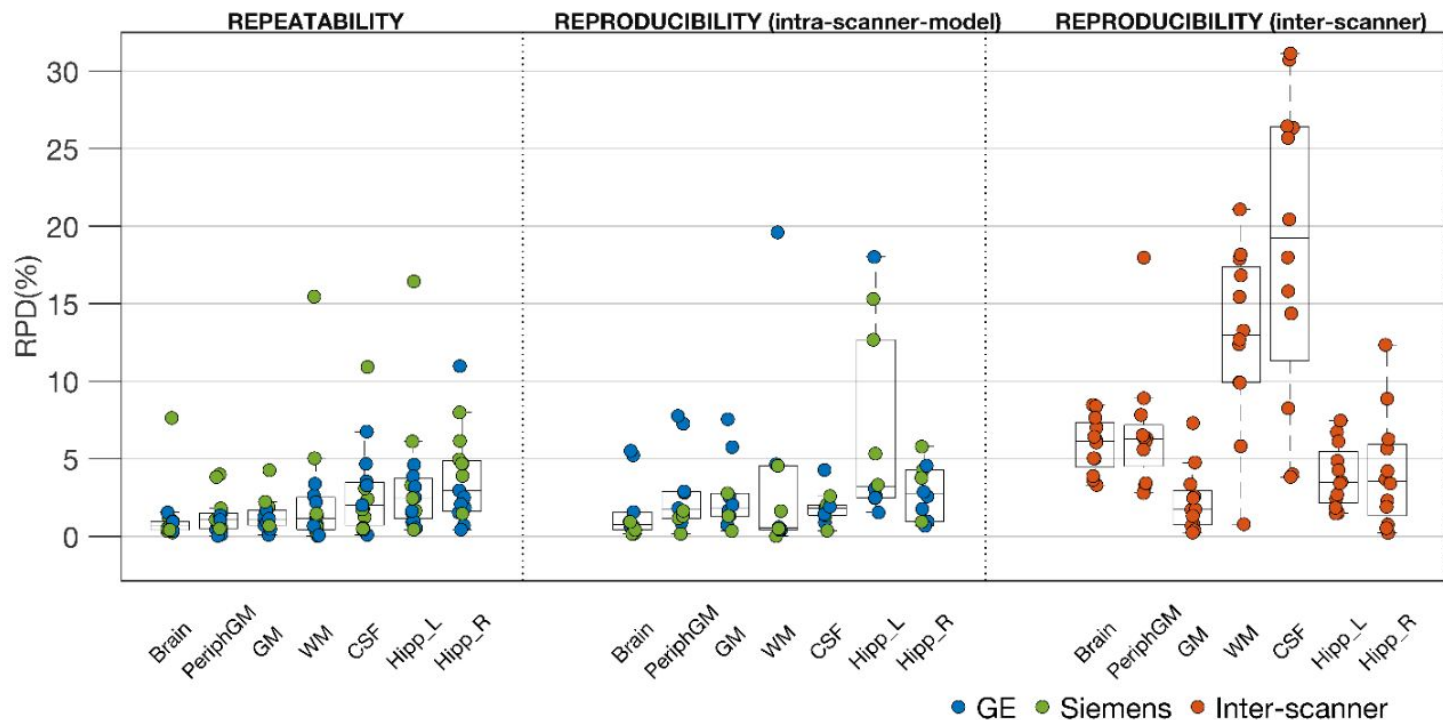
Automated QC classifier: Bhalerao et al., 2025, Imaging Neuroscience, doi: <https://doi.org/10.1162/IMAG.a.4>

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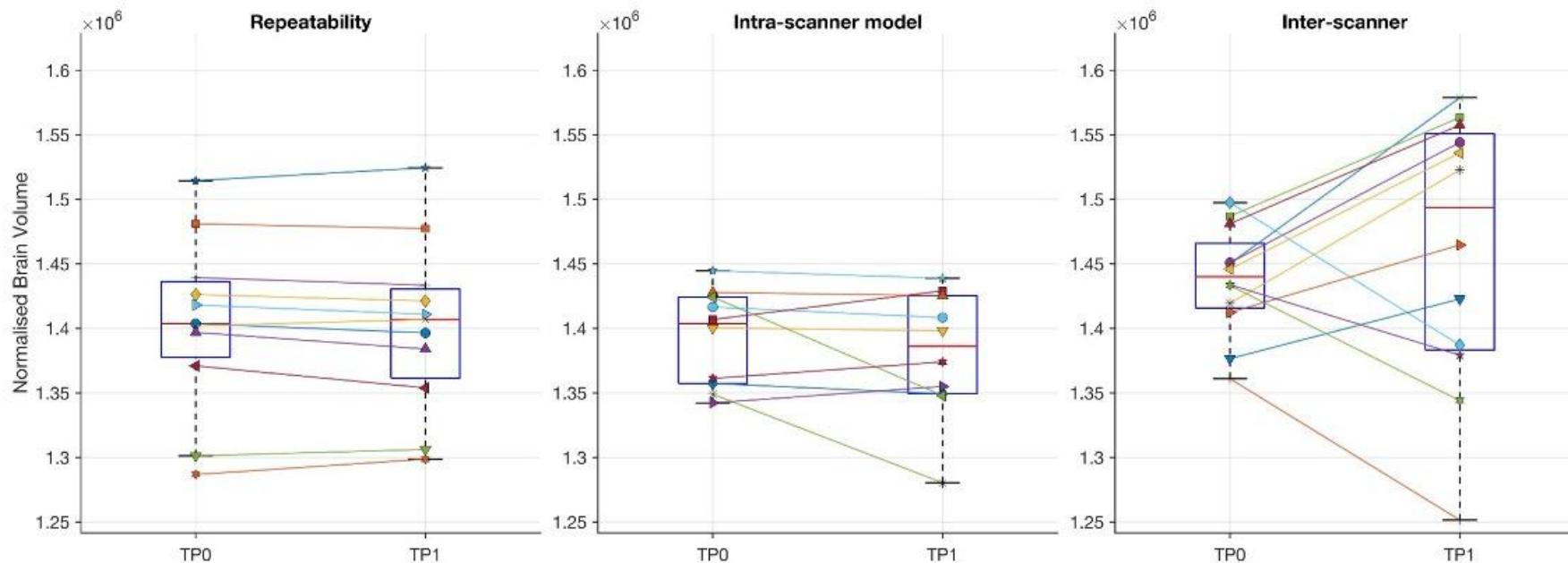
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Within-subject variability (no harmonization)



↑ Variability , ↓ Statistical power, ↑ Sample size required to detect true effects

Subject-order consistency (no harmonization)



↑ Variability , ↓ Statistical power, ↑ Sample size required to detect true effects

Harmonization evaluation

Before and after applying harmonization to your outcome measures (e.g., brain volumes, thickness, etc.) -

You must check -

- Site/batch effect before applying any harmonization
- If harmonization has removed the technical variability due to site/batch effect
- If harmonization has impacted the biological variability (e.g., age, sex, diagnosis, etc.)

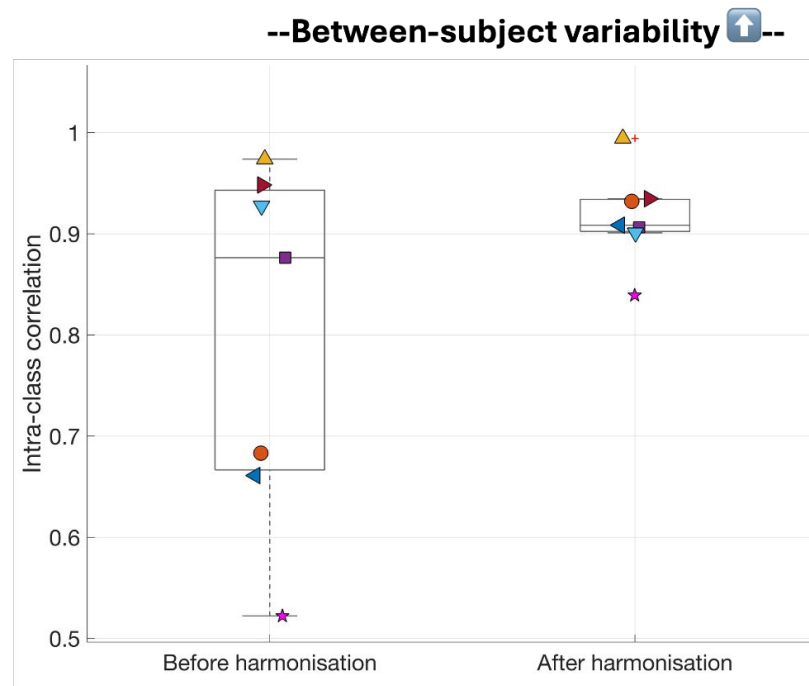
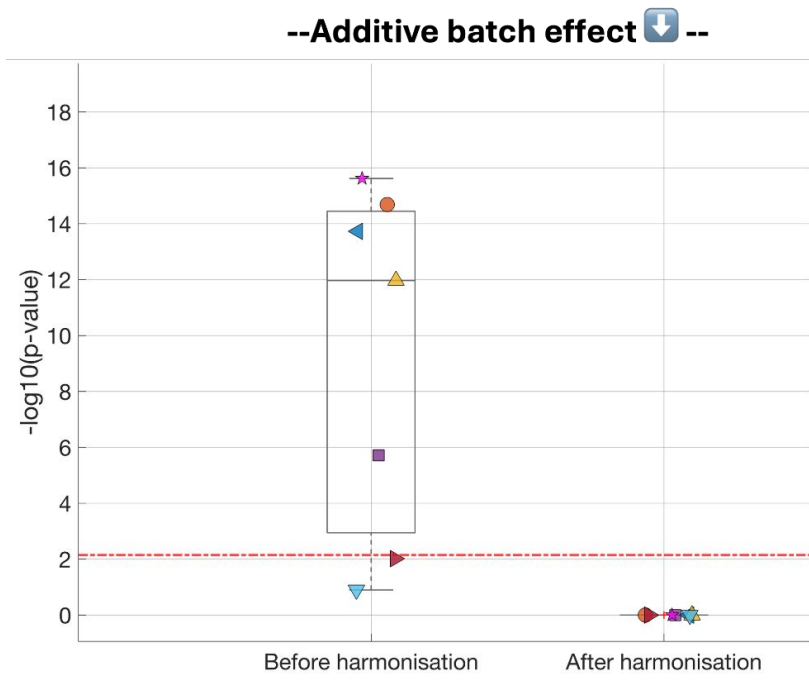


Harmonization Evaluation framework

- Removal of technical bias
 - Batch effects  - additive, multiplicative, multivariate
- Retain signal of interest
 - Biological variability   - between subject variability, association with biology (e.g., age)
- Stability of measures (appropriate for clinical/longitudinal test-retest)
 - Within-subject variability 
 - Subject order consistency  

Bhalerao et al. 2026, Harmonising Structural Brain MRI from Multiple Sites with Limited Sample Sizes,
doi: <https://doi.org/10.64898/2026.04.21.26351106>

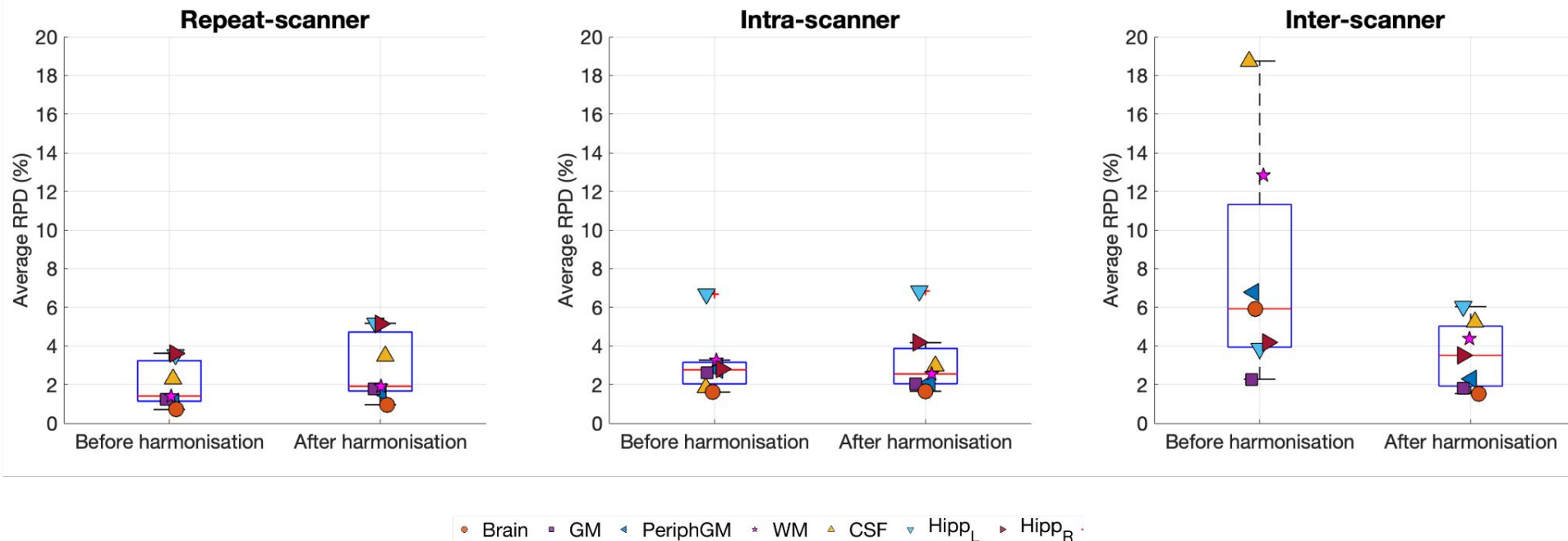
Harmonization Evaluation - Group-level



● Brain ■ GM ◀ PeriphGM * WM ▲ CSF ▼ Hipp_L ▶ Hipp_R --- p-value thr

Harmonization Evaluation - Within-subject variability

--Within-subject variability ↓--



In the notebook (B2 N1)

- Stability metrics -
 - Simulate data with known parameters
 - Calculate and plot within-subject variability and subject order consistency
- Group-level metrics -
 - Simulate data with known additive and multiplicative effects
 - Estimate and visualise additive and multiplicative batch effects



Multimetric evaluation (repeated measures)

<i>Effect</i>	<i>Metric</i>	<i>Pre-harmonisation</i>	<i>Post-harmonisation</i>
Within subject variability	Relative % difference or CoV	High values indicate poor reliability	Reduce
Subject order consistency	Spearman correlation	Low rank correlation indicates inconsistent subject ordering	Improve/preserve
Additive batch effects – If batch explains additional variance in data	<i>p</i> -values	Significance indicates presence of additive batch effect	Reduce
Pairwise batch effects	Number of significant site pairs	Significant differences indicate site-specific mean differences.	Reduce
Multiplicative batch effects – Variance comparison	<i>p</i> -values	Significance indicates presence of scaling batch effect	Reduce
Multivariate batch differences	Mahalanobis distances with overall mean and covariance structure	Large MD values indicate multivariate divergence of site distributions due to batch effects.	Reduce
Between-subject variability	Intra-class correlation = between-subject/total variability	Low ICC indicates dominance of residual or site-related variance over subject differences.	Improve/preserve
Biological variability – IDP association with e.g., age	Effect sizes, CIs, Significance	Attenuated or inconsistent associations may reflect site-related confounding or measurement noise.	Preserve/reveal true effects

Bhalerao et al. 2026, Harmonising Structural Brain MRI from Multiple Sites with Limited Sample Sizes, <https://doi.org/10.64898/2026.04.21.26351106>



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Multimetric evaluation (cross-sectional)

Effect:	Metric:	Pre-harmonisation:	Post-harmonisation:	Overview
IDP specific shifts	Cohen's d	Significant effect sizes seen in some or many IDPs	reduction of effect size magnitude	Remove covariate effects, compare scaled means
Global shifts	Mahalanobis distance	High overall distance	Reduced difference between batches and with data average	Compare average difference of means across all measures
IDP specific variation	F-test Linear modelling	Higher difference in variance between batches (far from 1) Explained variance by batch very high	Reduction of variance difference (closer to 1) Negligible difference between model fits when batch included	Variance difference between batch asses, linear model fits with and without batch assessed
Global difference	PCA analysis UMAP visualisation	Batch correlated with PC's, clear cluster separation UMAP visualisation shows global and non-linear differences	Low correlation in PC's, better correlation of covariates, no distinct batch clusters in either UMAP or PCA projection	Perform PCA, analyse variance structure across batches and show PCs by batch and covariates
IDP-IDP changes	Batch-Batch covariance difference	High difference between batch pairs in covariance structure	Reduction in normalised differences between covariance matrices	Estimate per batch covariance, take difference between batch pairs and scale by global covariance
Distributional differences	Two sample KS-test	Many features may show significant difference in distributions	Distributions more similar after harmonisation	Compare the empirical distribution of the batch residuals

<https://jake-turnbull.github.io/HarmonisationDiagnostics/>



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Video on Site regression and motivation for ComBat



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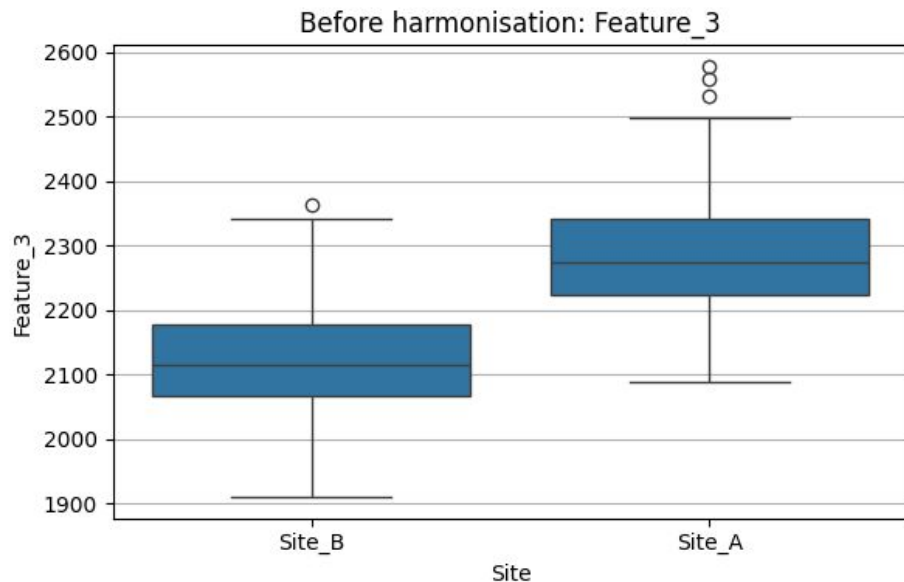
Site regression

- As a simple baseline, we can use regression to remove site-related mean shifts from the data.
- For each measurement, we fit a model like:
 - Observed value = Biological effects + Site effect + Random noise
 - Harmonized measurement = Observed value – site effect

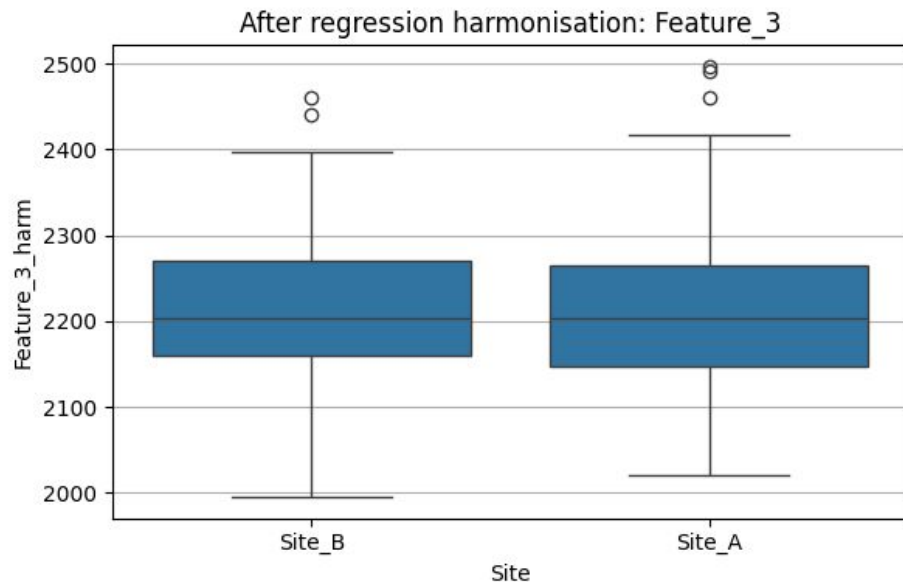


Site regression (before and after - case 1)

Site effects may show different means

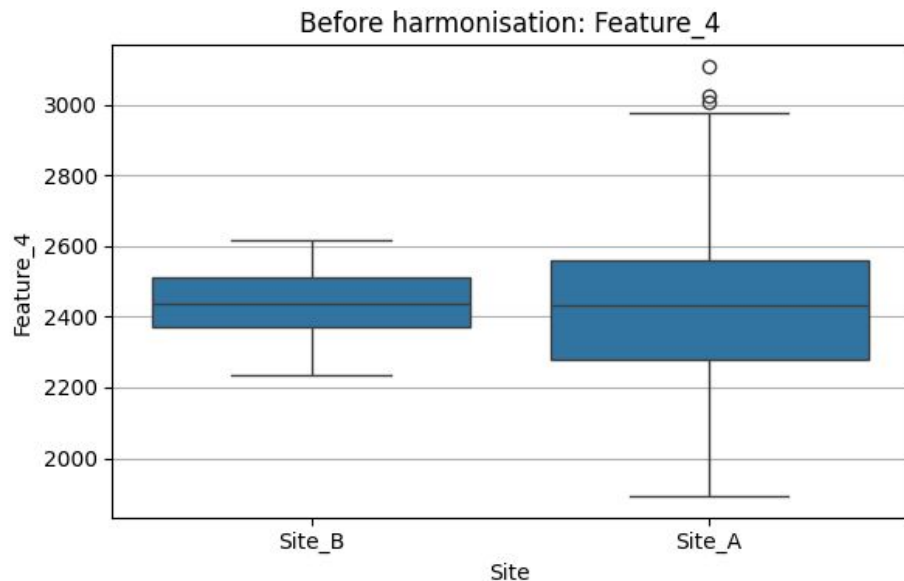


Regression shifts Site B to align with Site A.

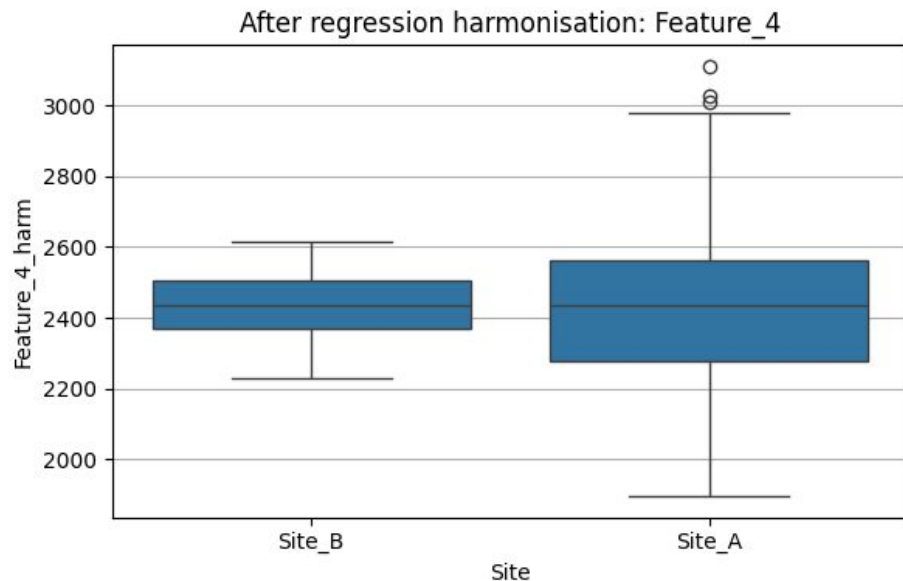


Site regression (before and after - case 2)

Site effects may show different spread



Site regression does not correct the spread.



Site regression (Notebook B2 N2)

1. Simulate longitudinal multi-site data
2. Visualize site-related effects
3. Apply regression-based harmonization
4. Visualize the harmonized data
5. Evaluate batch effects after harmonization



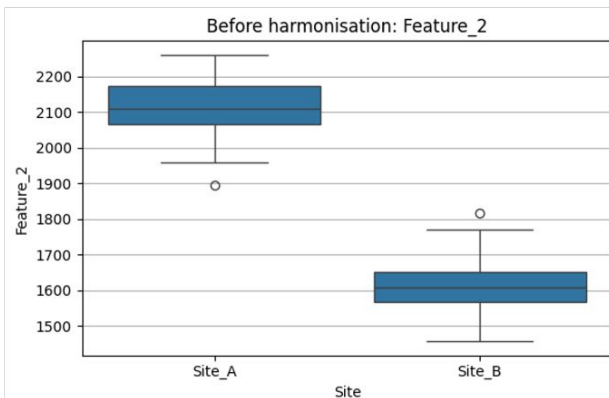
ComBat harmonisation (Basic intro)

Parameter	Cross sectional ComBat	Longitudinal ComBat
Removal of site effects	✓	✓
Preservation of biological effects	✓	✓
Accounts for repeated measures	X	✓
Models subject-specific baselines	X	✓
Conceptual representation	Observed value = Biology + Site effect	Observed value = Biology + Site effect + Subject effect
Data visual	Independent observations ○ ○ ○	Repeated measurements ○—○—○ ○—○—○ ○—○—○

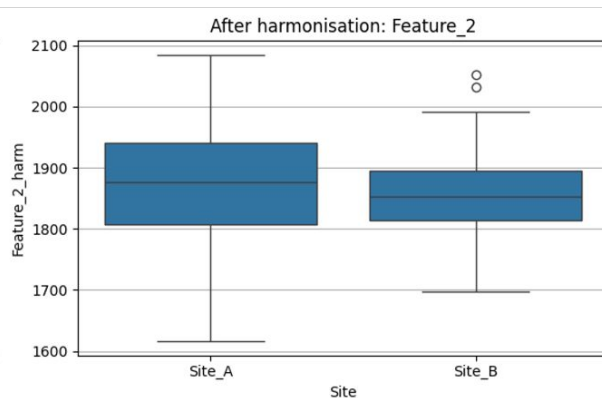


Site regression Vs. ComBat (Case 1)

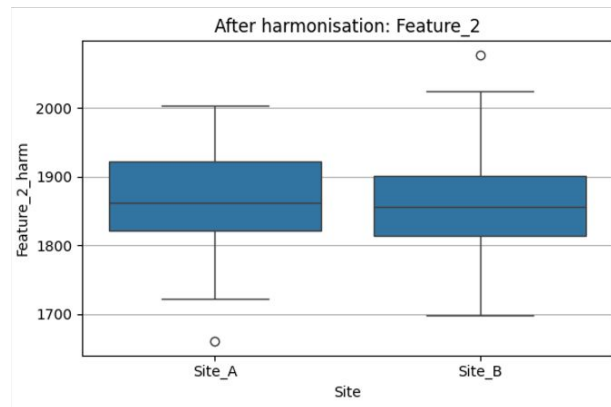
No harmonisation



Site regression



ComBat

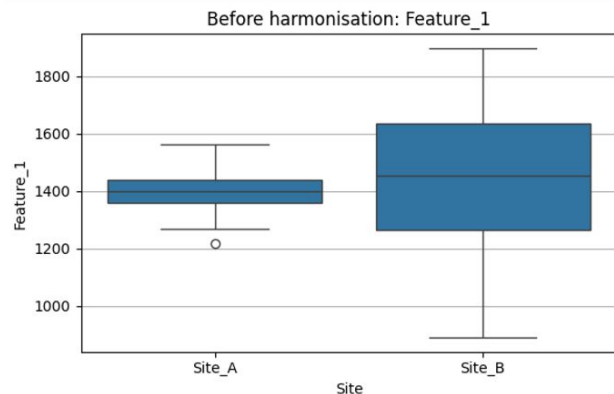


- Before harmonisation: Site-related differences in both location and scale are clearly visible.
- After site regression: The location difference is removed, but scale differences persist.
- After ComBat harmonisation: Both location and scale differences are substantially reduced.

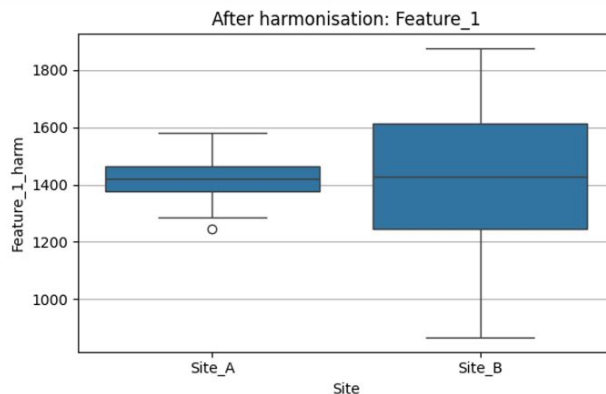


Site regression Vs. ComBat (Case 2)

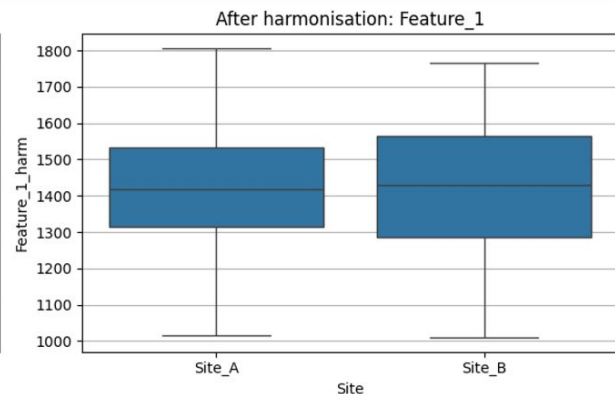
No harmonisation



Site regression



ComBat



- Before harmonisation: Similar central tendency across sites, but marked differences in variability.
- After site regression: Minimal change, reflecting the absence of a site-related mean shift.
- After ComBat harmonisation: Variability was aligned across sites, improving distributional comparability.



Site regression Vs. ComBat (Notebook - B2 N3)

- Simulate data across multiple sites.
- Introduce additive and multiplicative site/batch effects
- Apply regression-based harmonisation.
- Visualize the data before and after regression correction.
- Apply ComBat harmonisation.
- Compare the harmonized results using visualizations and statistical evaluation.



BLOCK 3



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ComBat for the harmonisation of derived measures

ComBat is the most frequently used harmonisation method for MRI derived features

Originally ComBat (Combatting batch effects) was used to mitigate differences arising when microarray gene expression data was produced by different locations, people or methods

ComBat was adapted for MRI derived features in 2017 by Fortin et al.,

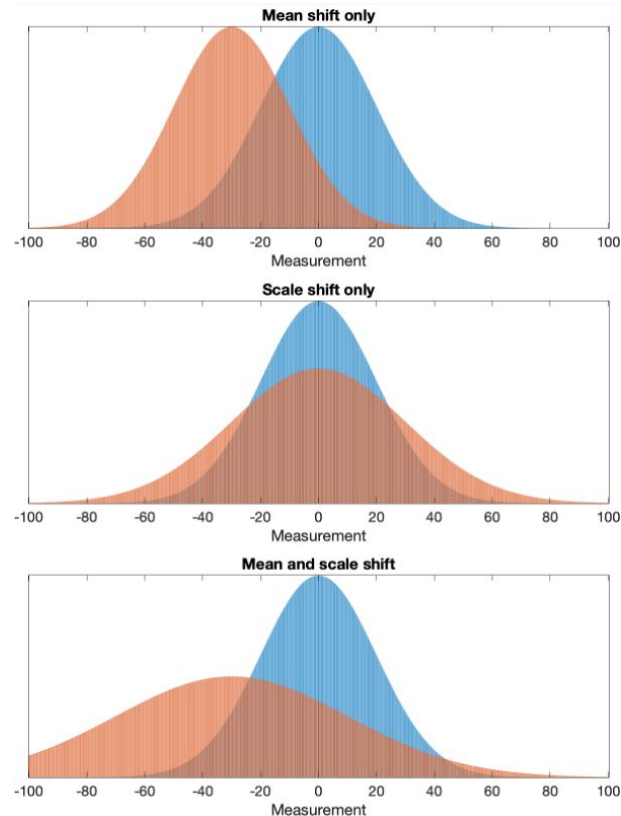


ComBat for the harmonisation of derived measures

ComBat is an empirical-Bayes, location and scale correction method for adjusting batch effects

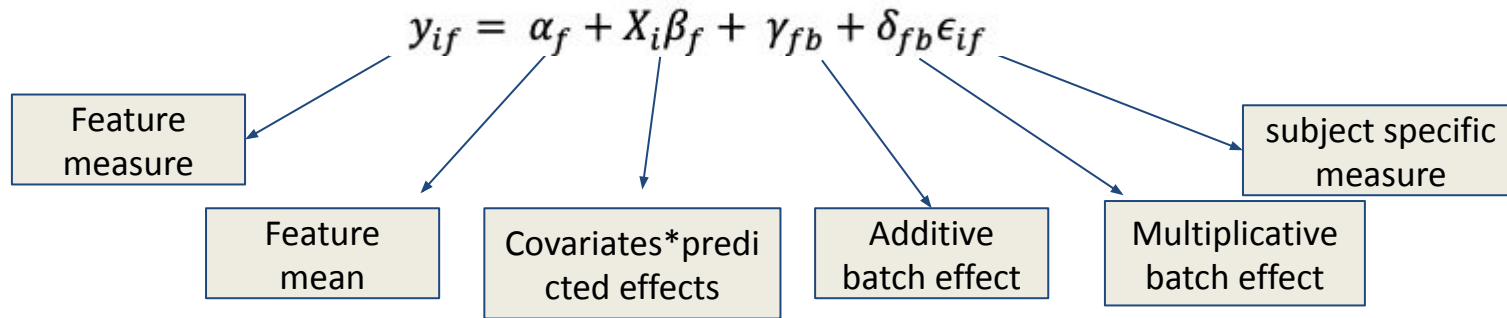
- Location effects refer to differences in the mean
- Scale effects to differences in the variance/spread of the data

Can be beneficial over regression when there is a large difference in the variance between batches and when sample size is small



ComBat for the harmonisation of derived measures

ComBat assumes that data is of the form:



Before estimating the batch effects, ComBat first standardises the data so that it is mean centered, unitless and covariate effects are removed

$$Z_{if} = \frac{y_{ifb} - \alpha_f - X_i\beta_f}{\sigma_f} = \frac{\gamma_{fb} + \delta_{fb}\epsilon_{if}}{\sigma_f}$$

The batch effects are then estimated using an iterative empirical-Bayes (EB) algorithm

Starting from a reasonable point estimate for the batch location and scale (i.e the beta from OLS on the residuals and the batch variance), get a new value for the estimates of gamma and delta and repeat until convergence

$$Y_{if}^* = \frac{\sigma_f}{\hat{\delta}_{fb}^*} (Z_{if} - \hat{\gamma}_{fb}^*) + \alpha_f + X_i\beta_f$$

ComBats strengths lie in its ability to stabilise estimated corrections using data driven priors

ComBat looks at the batch characteristics across features and uses them to construct two data driven fixed priors of the estimated batch effect for each batch of data:

$$\gamma_{sf} \sim (\gamma_s, \tau_s^2) \text{ and } \delta_{sf}^2 \sim \Gamma(\lambda_s, \theta_s)$$

The hyperparameters here are data driven and shrink estimates of the correction when data size is small.

The iterative process then balances the feature specific likelihood against the batch specific prior:

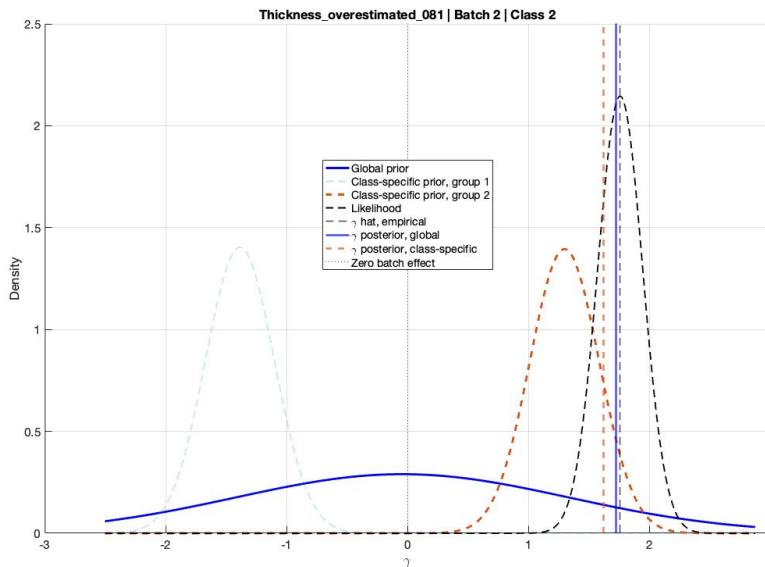
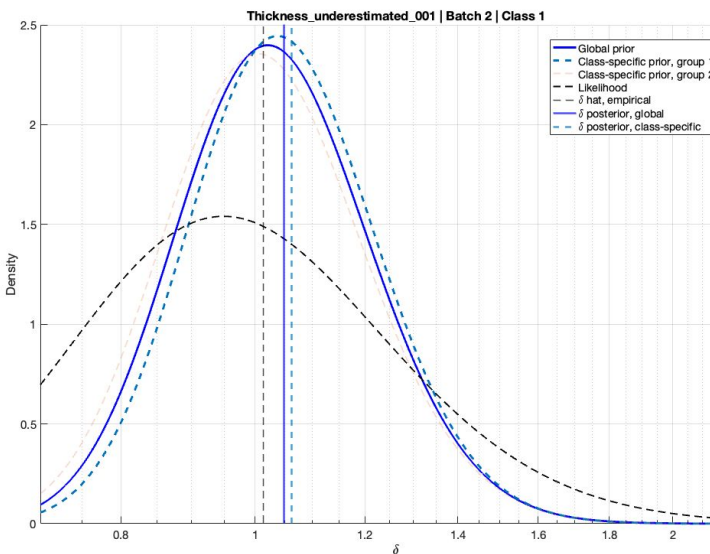
More data: stronger likelihood dominance, weaker shrinkage

Less data: More reliance on the prior, stronger shrinkage



For prior performance to be beneficial, batch effects need to be relatively **homogenous** within a dataset: ComBat **CAN** be ran on more than one type of measure at once, but this is often ill informed:

Take for example a dataset where cortical thickness is underestimated in one region, but overestimated in another, we can see that the prior has a relatively small influence on the estimate when applied to all features as opposed to only ones with the same direction of bias:



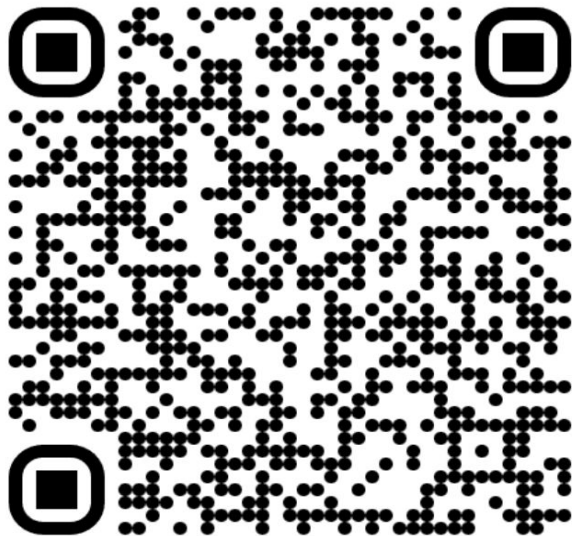
Better prior
-> More EB
shrinkage

Worse prior
-> Heavier
reliance on
the first
estimate



Applying ComBat

Some examples of applying ComBat for mitigating batch/site effects is given in the notebooks.



Several versions are available online in multiple coding languages. We discuss these and provide a version within the notebook.

[notebooks/block03/B3_N0_ComBat_applications.ipynb](#)



ComBat Summary

ComBat can be incredibly powerful for correcting location and scale effects when used correctly

ComBat should only be applied where there is a sufficient sample size (~ 30) in each batch

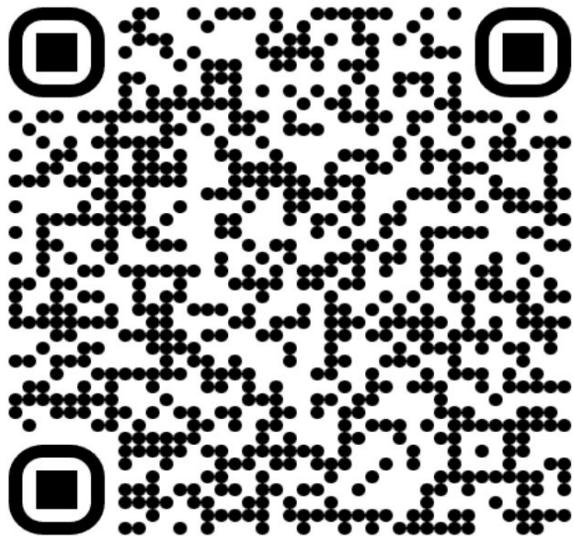
Imbalance in batch sizes can be harmful if not intended, as larger batches will dominate the predicted mean, variance and covariate effects, making the predicted corrections to larger batches weaker than smaller batches

ComBats EB shrinkage works best if features with homogenous batch effects are grouped together, otherwise the likelihood of the first feature specific estimate will dominate, with this being most important for scenarios where data is sparse



ComBat Limitations - Notebook

As good as it is, ComBat has some important limitations which we will now look at in the [notebook](#).



notebooks/block03/B3_N1_ComBat_limitations.ipynb



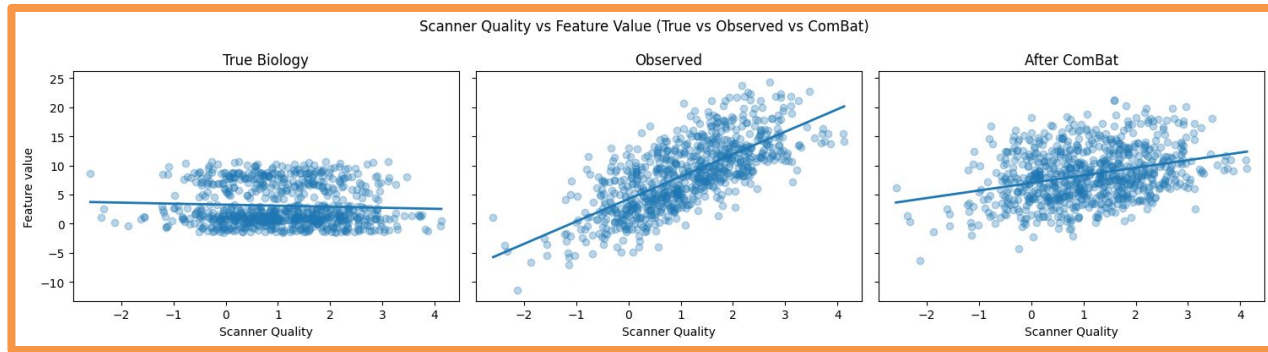
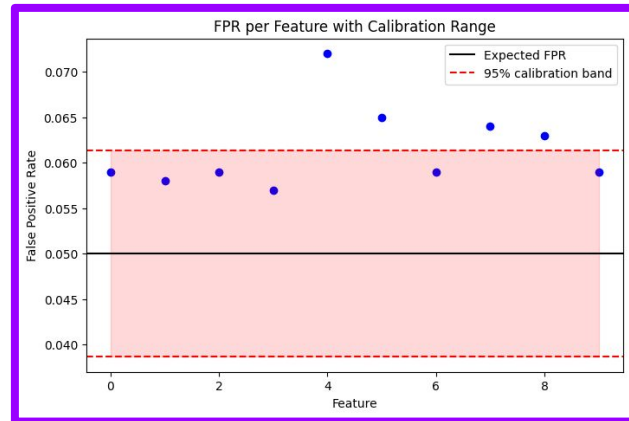
ComBat Limitations - Summary

- In the notebook we have seen some modeling limitations:
 - Linear structure;
 - Restricted to additive and multiplicative (location–scale) adjustments;
 - limited ability to model complex interactions between biology and scanner;
 - Discrete scanner/site grouping.



ComBat Limitations - Summary

- Which have important statistical consequences:
 - **False positive rate (FPR) inflation;**
 - Potential removal/distortion of true biological signal when assumptions are violated;
 - **Incomplete removal of scanner effects.**



ComBat Limitations - Summary

There are also some practical limitations:

- Requires sufficient sample size per scanner for stable estimation
- Not applicable when: scanner metadata is missing, anonymized, or unreliable

These are fundamental limitations of location–scale harmonization models, not specific to ComBat alone.



Coffee break



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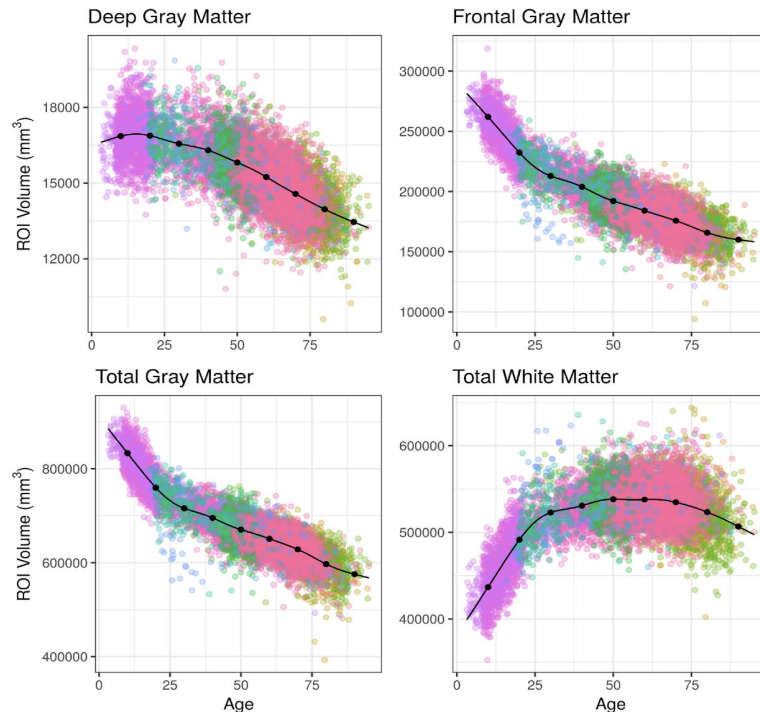
Alternatives to ComBat

Several conceptual extensions to ComBat have been developed to broaden its applicability and address some of these limitations



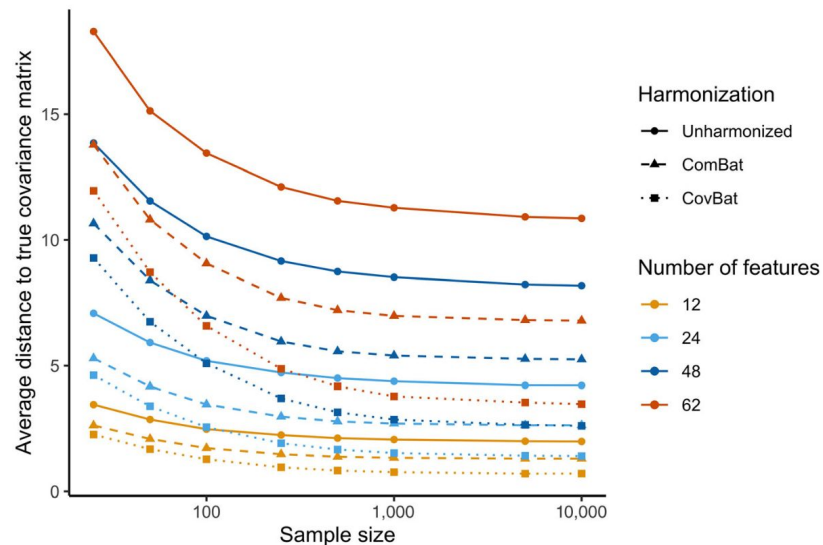
ComBat-GAM

- Replaces linear covariate effects with smooth functions using **generalized additive models (GAMs)**
- Uses penalized splines to estimate nonlinear biological effects
- Site effects are still:
 - discrete (site labels)
 - additive and parametric (location–scale)
- No modelling of complex scanner–covariate interactions



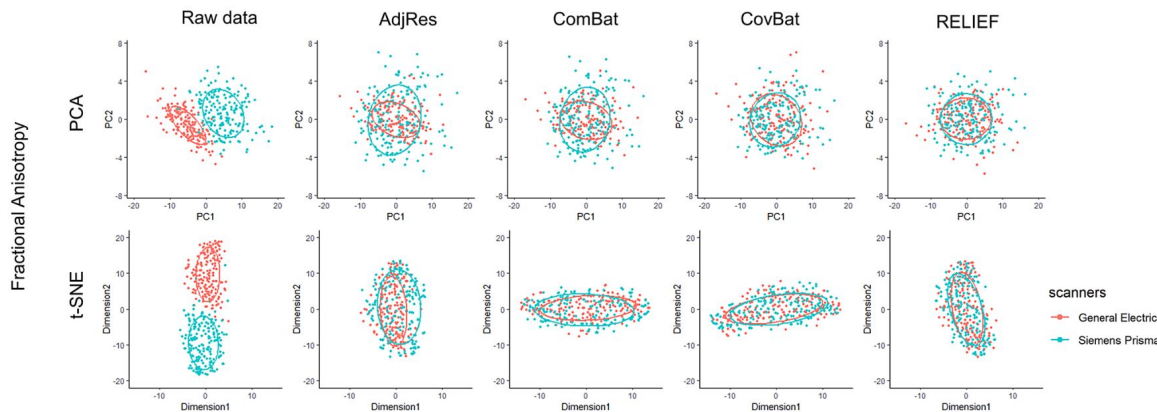
CovBat

- CovBat performs standard ComBat to remove mean/variance site effects.
- Then applies PCA on residuals and harmonizes PC scores across sites using ComBat again
- Reconstructing data yields adjusted covariance structure
- However:
 - linear / Gaussian assumptions
 - depends on site labels
 - Only adjusts low-rank covariance



RELIEF

- RELIEF decomposes data into a shared structure across sites (which is preserved) and site-specific latent effects (which are removed)
- Uses low-rank matrix factorization to isolate scanner-related variation
- However, still relies on:
 - explicit site grouping
 - linear latent structure



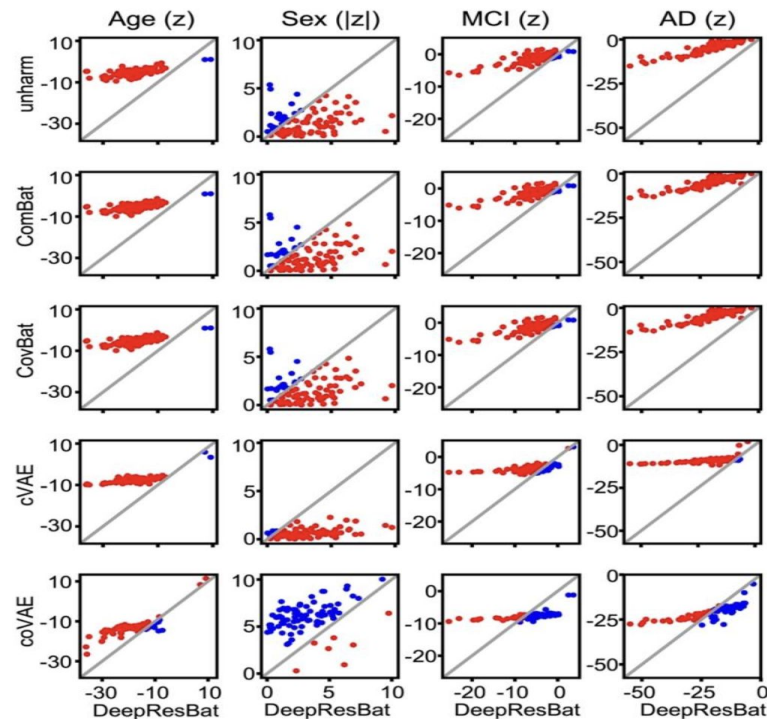
ComBatLS

- Extends ComBat to allow variance to depend on biological covariates
- Uses a location–scale model:
 - mean depends on covariates
 - variance also depends on covariates
 - Applies EB shrinkage to site-specific variance scaling factors
- Still:
 - parametric
 - additive site effects
 - requires site labels



DeepResBat

- DeepResBat is a two-stage hybrid model:
 - Regression + ComBat
 - Train deep neural network on residuals
- Network learns nonlinear site effects conditioned on covariates and removes predicted site signal from residuals
- However,
 - Not a unified statistical model but a sequential pipeline
 - Still relies on discrete scanner IDs
 - Risk of overfitting and/or removing biological signal unintentionally



Alternatives to ComBat

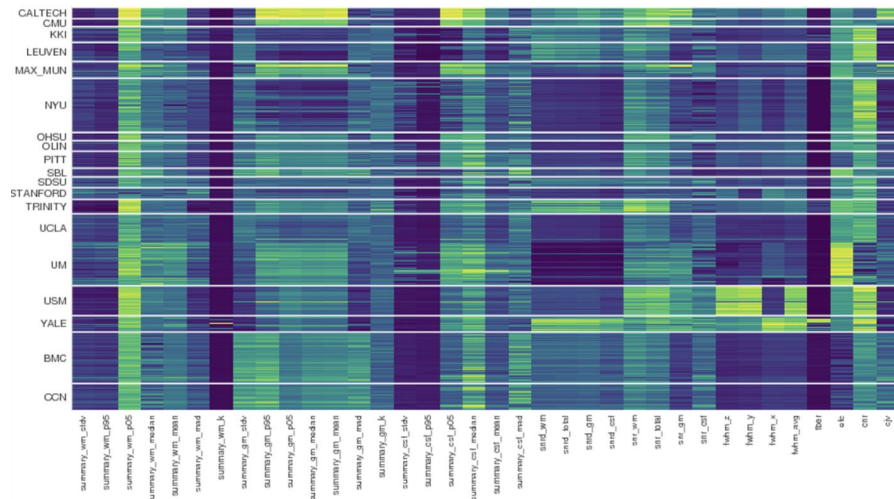
- **ComBat-GAM** → fixes *mean structure (nonlinearity)*
- **ComBatLS** → fixes *variance structure*
- **CovBat / RELIEF** → fixes *multivariate structure*
- **DeepResBat** → tries to fix *nonlinearity + interactions*

- But none of them:
 - move beyond **discrete site modelling**
 - or fully abandon **parametric assumptions**



Introducing IQMs

- Image Quality Metrics (IQMs) quantify intrinsic image properties
- Studies have shown that **IQMs are strongly associated with MRI scanners and acquisition protocols**
- Importantly, IQMs can be computed directly from individual MRI scans without requiring knowledge of scanner identity.



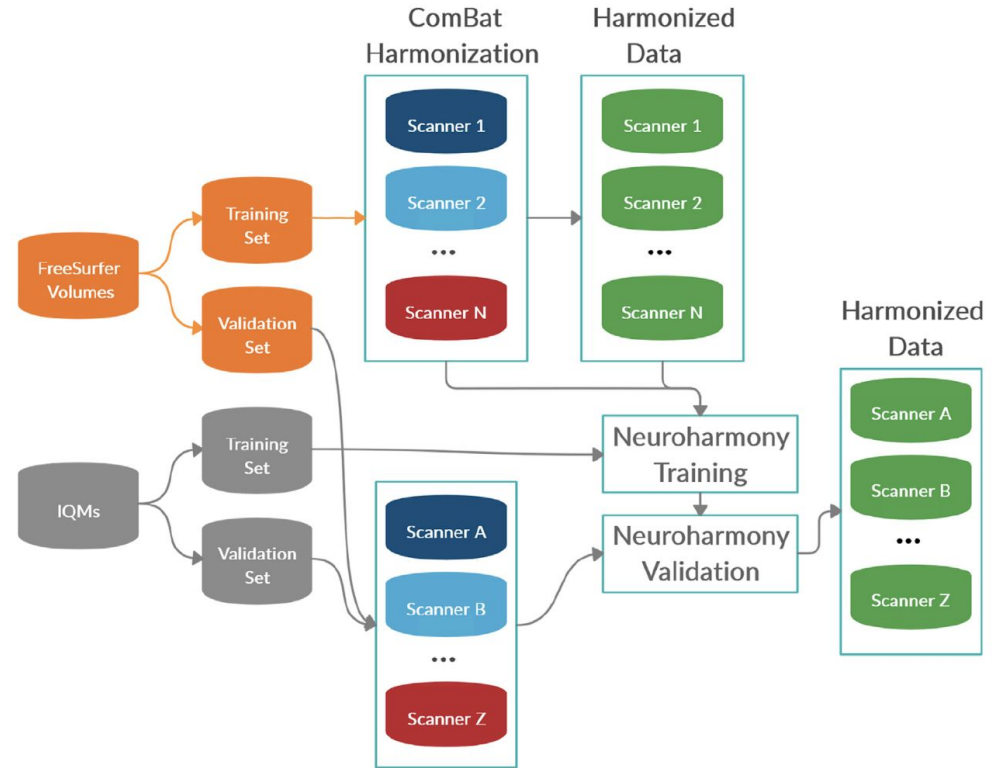
Neuroharmony

- Neuroharmony represents a notable attempt to incorporate IQMs into the harmonization process.
- Two-stage harmonization approach:
 - Stage 1:
 - Apply ComBat on training data with known scanner labels
 - Estimate additive and multiplicative batch effects for each feature
 - Stage 2:
 - Train machine learning model to predict ComBat-derived corrections from IQMs
- At deployment predict scanner-related corrections using IQMs only



Neuroharmony

- However,
 - It only learns to approximate ComBat corrections from IQMs
 - It inherits ComBat assumptions (and main limitations)
 - It's not a new harmonization framework.



BARTTharm: IQM-based Harmonization

- BARTTharm is a fully data-driven statistical harmonization framework built on Bayesian Additive Regression Trees (BART)
- Uses IQMs as continuous covariates to capture acquisition-related variability, scanner IDs can be included, but are not required
- Jointly estimates:
 - biological effects (from covariates)
 - scanner effects (from IQMs)
- Both components are learned simultaneously within a unified Bayesian model instead of multi-stage pipelines
- However, BART-based inference (MCMC) is computationally intensive

BARTTharm: MRI Harmonization Using Image Quality Metrics and Bayesian Non-parametric. Emma Prevot, Dieter A. Häring, Laura Gaetano, Russell T. Shinohara, Chris C. Holmes, Thomas E. Nichols, Habib Ganjgahi bioRxiv 2025.06.04.657792; doi: <https://doi.org/10.1101/2025.06.04.657792>



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BARTharm: IQM-based Harmonization

- BARTharm addresses main limitations of LS models
- It captures non-linear effects, high-order interactions, complex covariate dependencies avoiding parametric LS assumptions, and without requiring scanner IDs

Experiment	RMSE		BIO Corr		IQMs Corr		FPR	
	BARTh	ComB	BARTh	ComB	BARTh	ComB	BARTh	ComB
Misspecification	2.17	2.25	0.95	0.78	0.03	0.23	0.056	0.055
0.30 Correlation	0.87	2.98	0.99	0.81	0.12	0.47	0.056	0.075
0.50 Correlation	1.77	2.87	0.99	0.90	0.03	0.30	0.058	0.088
0.75 Correlation	2.67	3.14	0.99	0.92	0.04	0.15	0.057	0.072

τ (BIO)	μ (IQMs)	RMSE		BIO Corr		IQMs Corr		FPR	
		BARTh	ComB	BARTh	ComB	BARTh	ComB	BARTh	ComB
Linear	Linear	0.21	0.24	0.998	0.953	0.007	0.174	0.051	0.055
Linear	Non-Linear	0.65	1.07	0.999	0.381	0.001	0.816	0.044	0.062
Non-Linear	Linear	0.24	0.53	0.999	0.741	0.003	0.421	0.049	0.045
Non-Linear	Non-Linear	0.23	0.52	0.999	0.740	0.003	0.454	0.047	0.063



And what about Longitudinal Data?

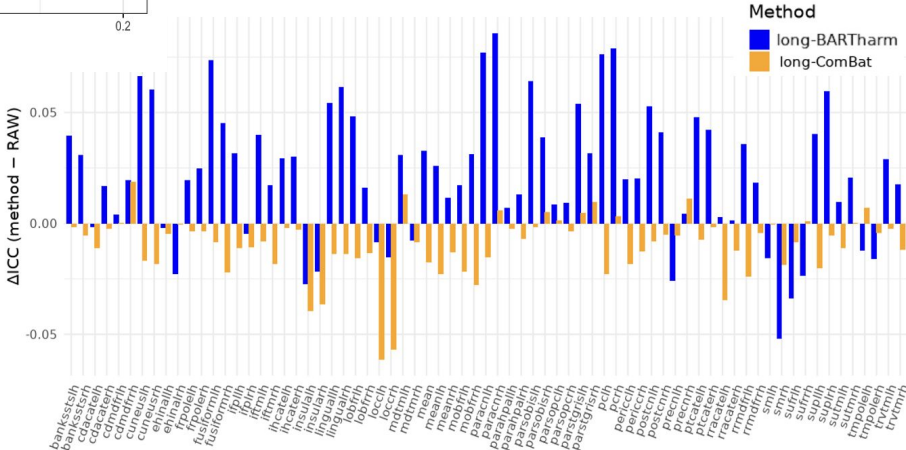
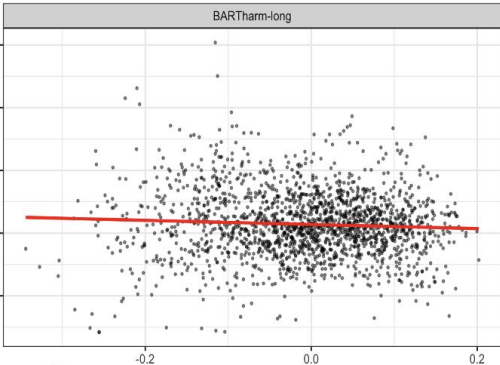
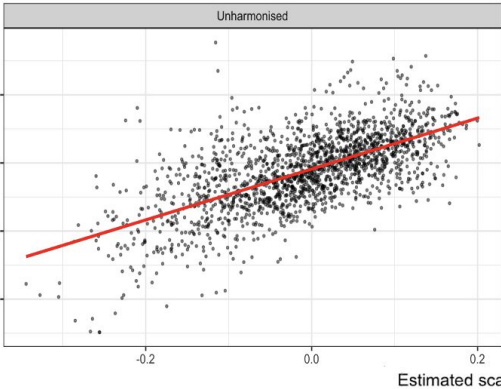


BARTharm-long

- We have seen long-Combat before. However:
 - It assumes a simple random-intercept-only structure and
 - It inherits key limitations of the ComBat framework (reliance on linear additive models and dependence on scanner/site ID).
- **long-BARTharm** is the first longitudinal, non-linear, IQM-based harmonization method.
- It extends BARTharm incorporating subject-specific random intercepts and slopes, explicitly modelling within-subject correlation and accommodating diverse longitudinal trajectories.



BARTTharm-long



Data harmonization in Machine Learning



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ML hypothesis and assumptions.

Predicting on new unseen data.

- ML models are designed to predict on future unseen data.

More data, better models.

- When we have more data, our ML models can perform better.



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ML hypothesis and assumptions.

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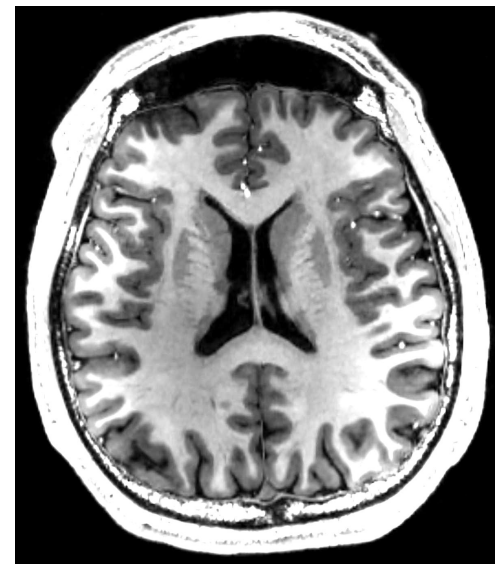
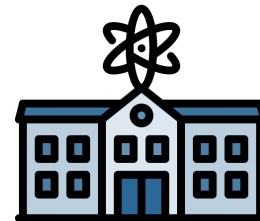
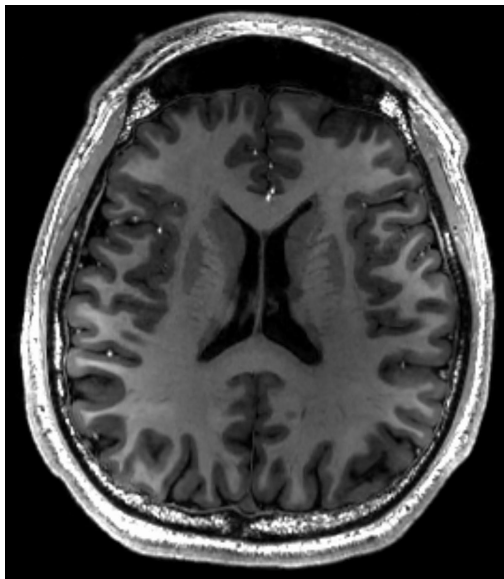
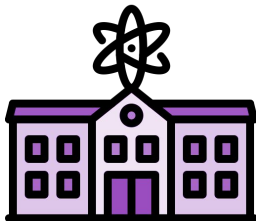
Independent and identically distributed.

- The data is IID, which we know is not the case for multisite data.

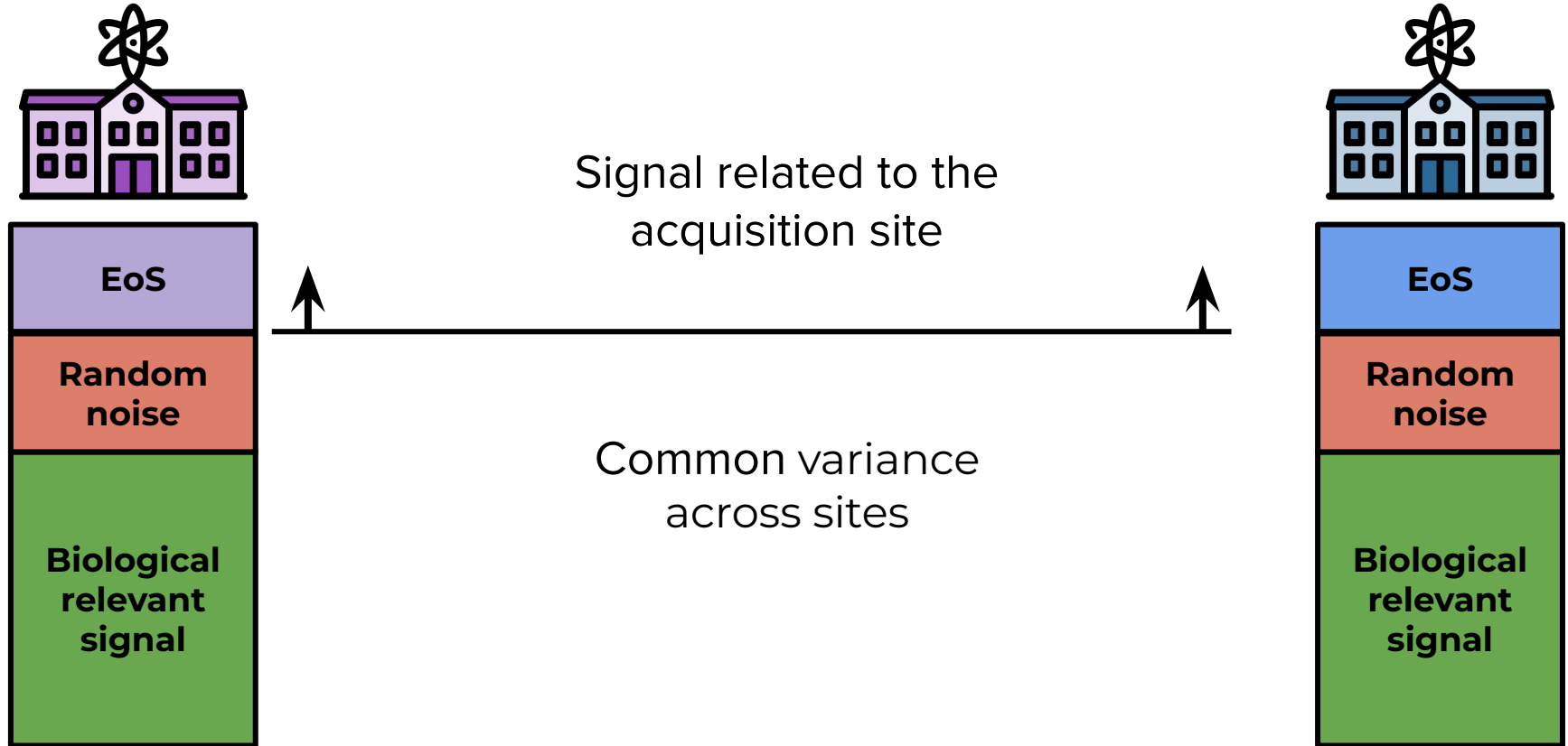
Harmonization enters in the game to help in 3 and improve 2, but we need to be careful with 1.



Effect of Site (EoS)



Effect of Site (EoS)



EoS as confound

Site acts as a confound

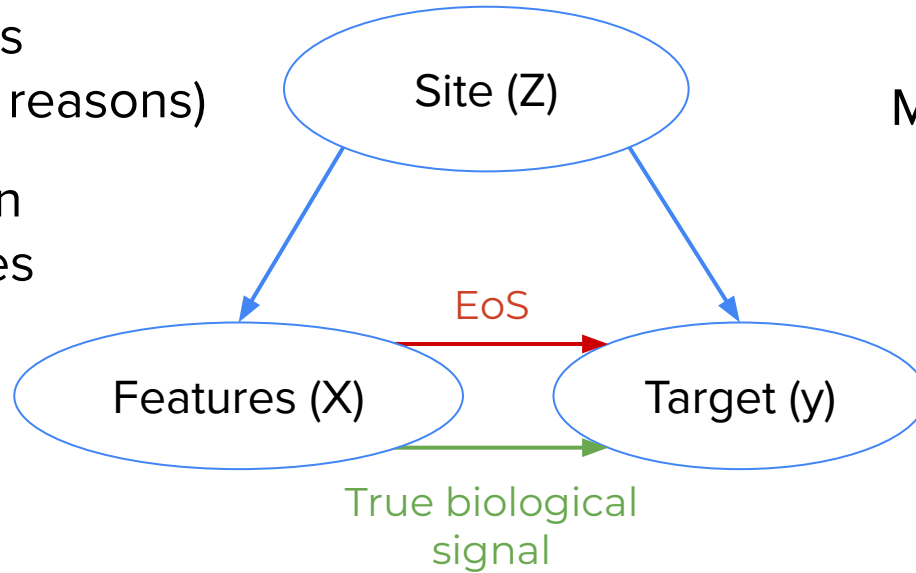
Measurement differences
(Technical reasons)

Population differences

Measurement differences
(Technical reasons)

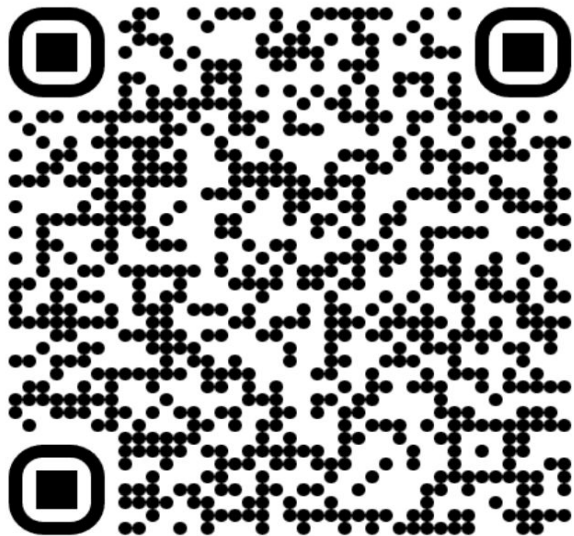
Measurement differences
(Target definition)

Population differences
(Acquisition imbalance)



How will the EoS affect the ML performance?

Will the performance drop or increase when we have EoS?



`notebooks/block03/B3_N2_EoS_in_ML.ipynb`



Site-target (in)dependence

Site-target dependence happens when different class proportion were acquired in each sites, e.g. **all control are acquired in one site and all patients on a different site.**

Site-target dependence



Site A

Site B

Site-target independence



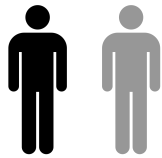
Site A

Site B

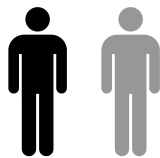


Site-target independence

Site-target independence

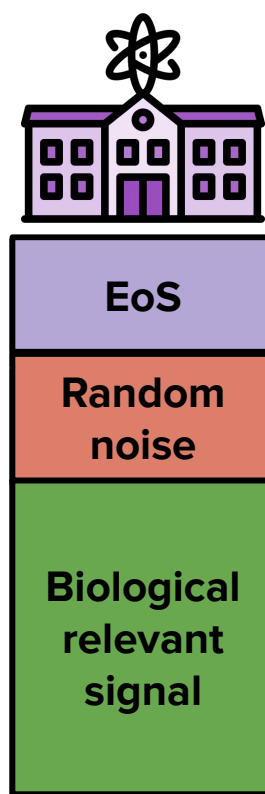


Site A

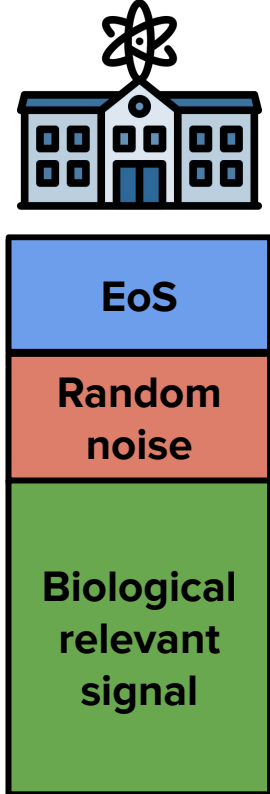


Site B

Overall balance: 0.5



EoS acts as Noise



Site-target dependence

Site-target dependence

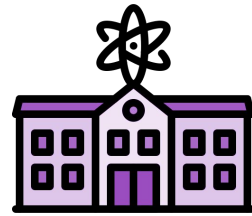


Site A



Site B

Overall balance: 0.5



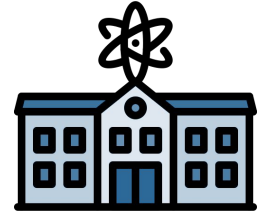
Random
noise

EoS

Biological
relevant
signal

EoS acts as “signal”

The signal and
site are
confounded!



Random
noise

EoS

Biological
relevant
signal

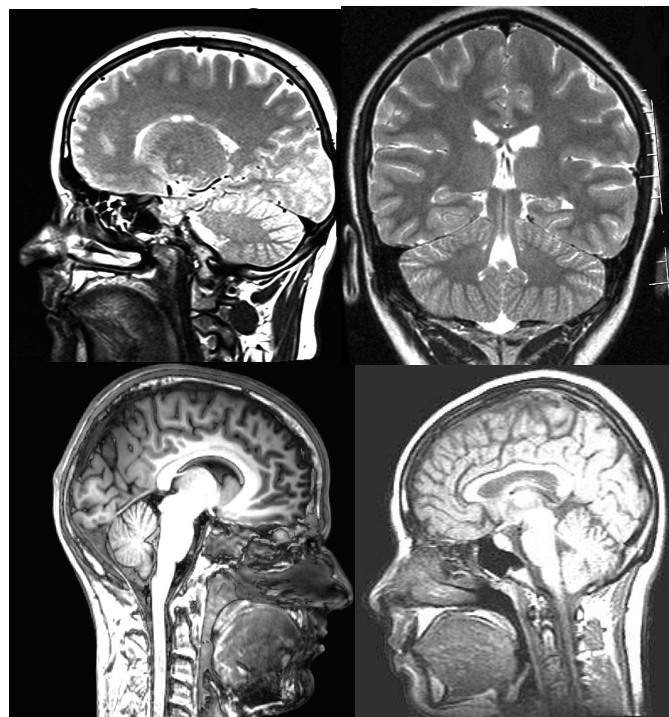


Site-target dependence

Control



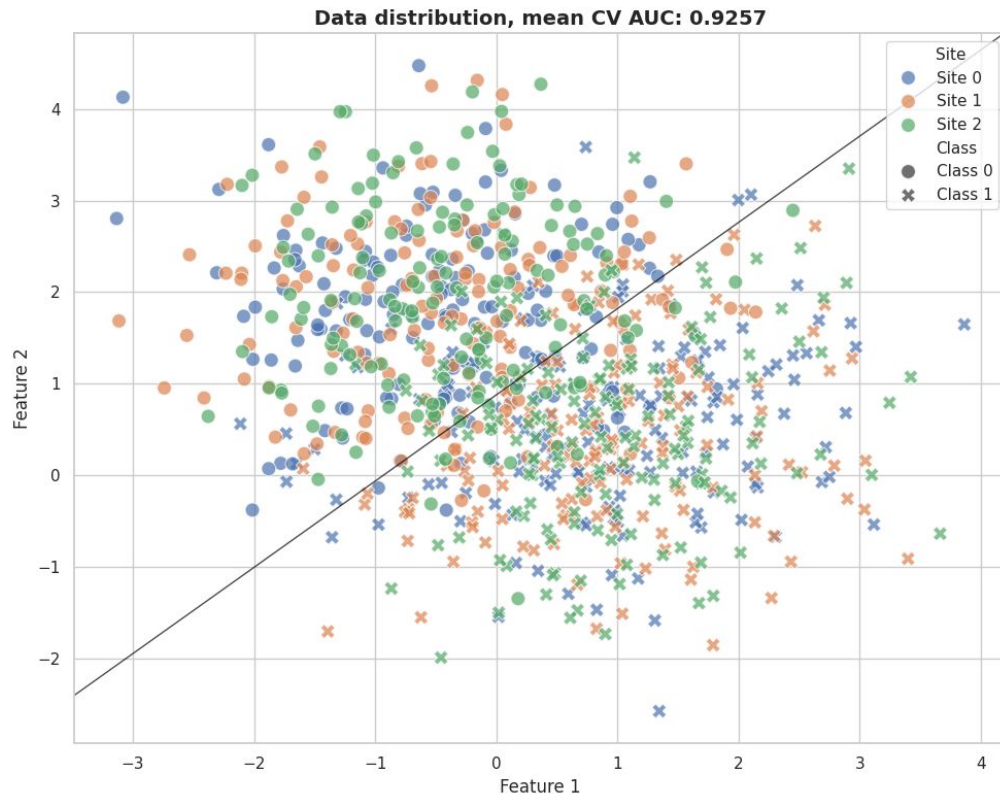
Patient



How will the EoS affect the ML performance?

Problem characteristic:

- Real signal: Yes
- Effect of site: No
- site-class imbalance: No

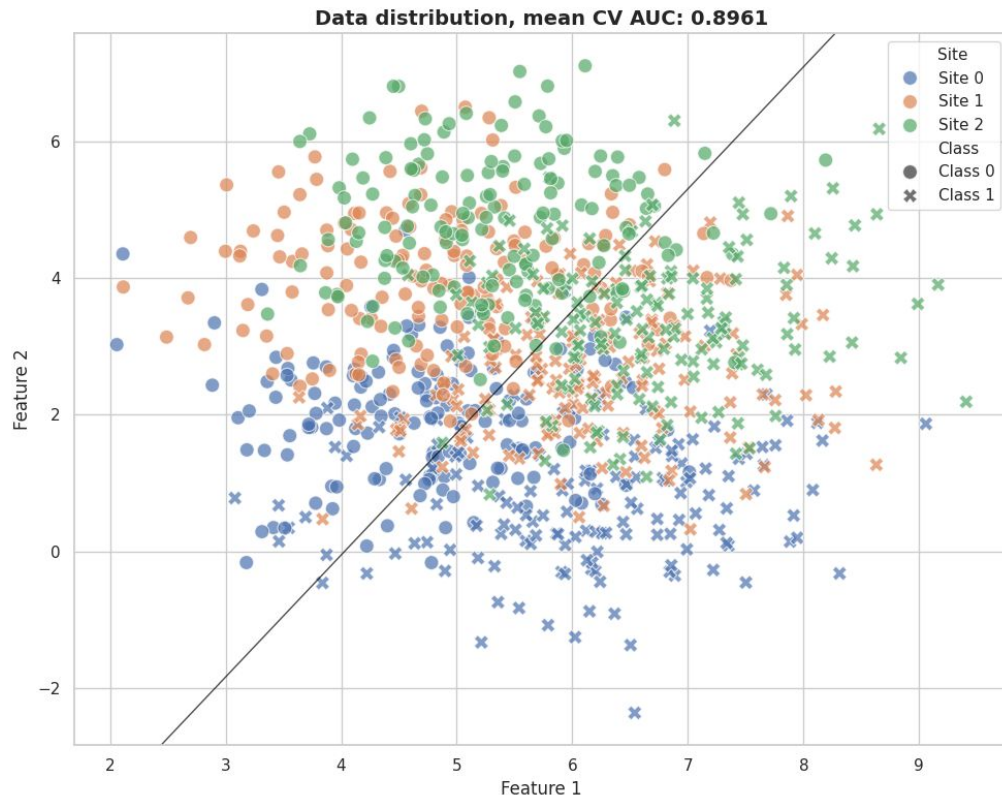


How will the EoS affect the ML performance?

Problem characteristic:

- Real signal: Yes
- Effect of site: Yes
- site-class imbalance: No

EoS as noise,
lower performance

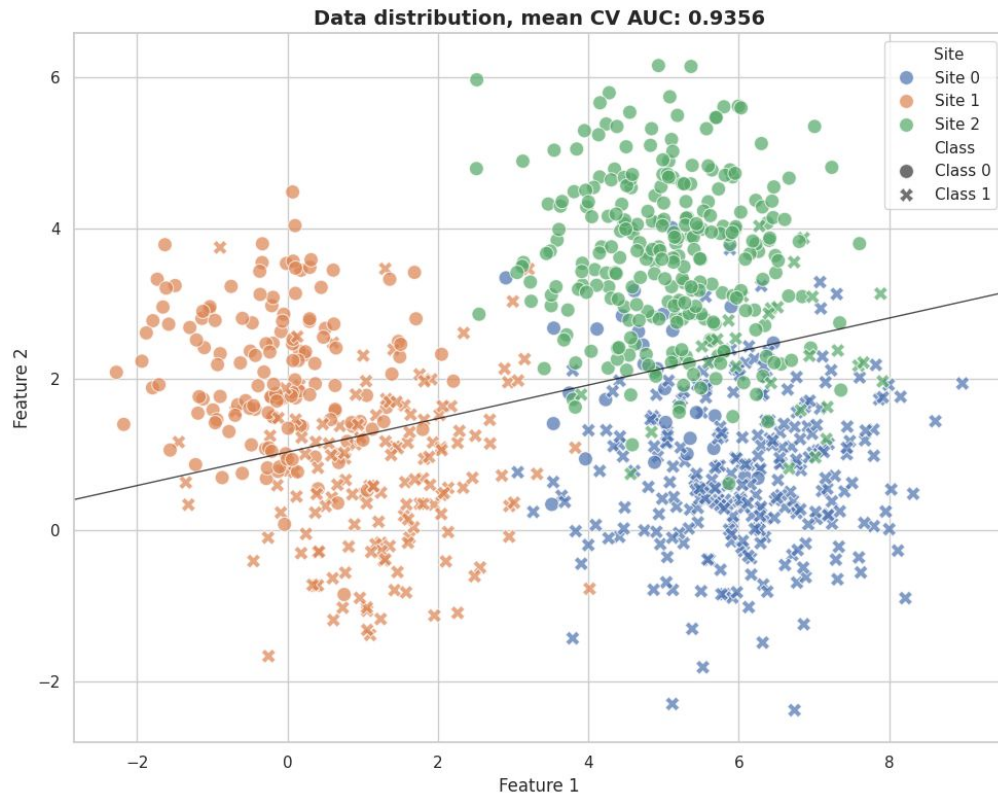


How will the EoS affect the ML performance?

Problem characteristic:

- Real signal: Yes
- Effect of site: Yes
- Site-class imbalance: Yes

EoS as “signal”,
higher performance.

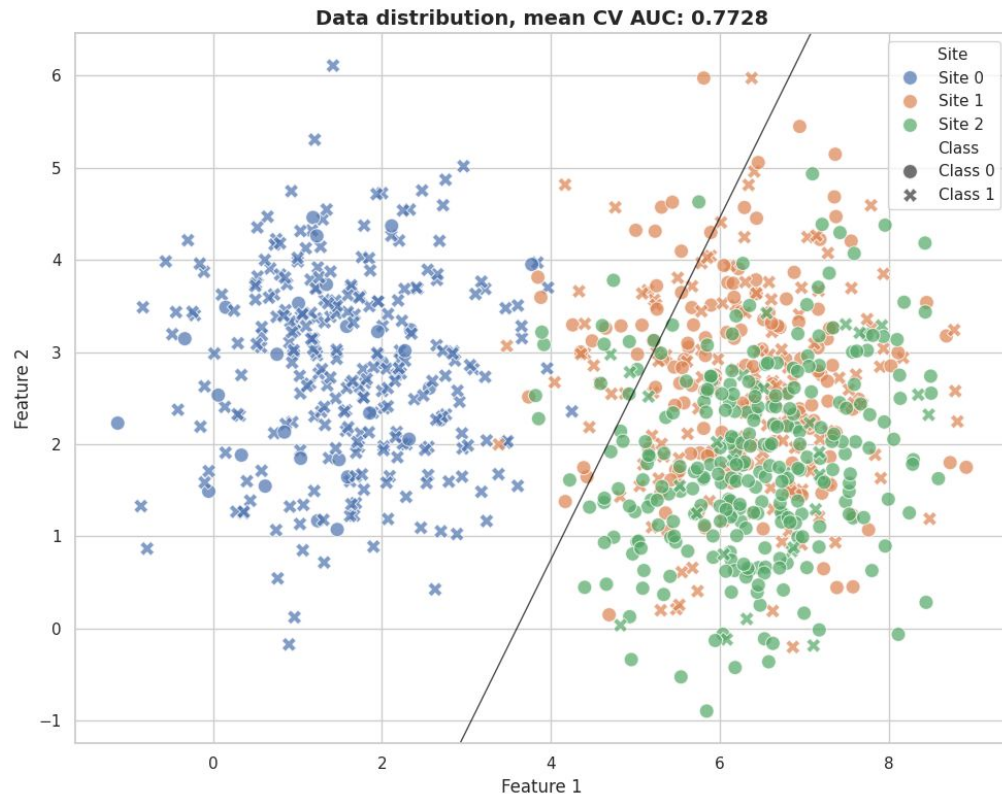


How will the EoS affect the ML performance?

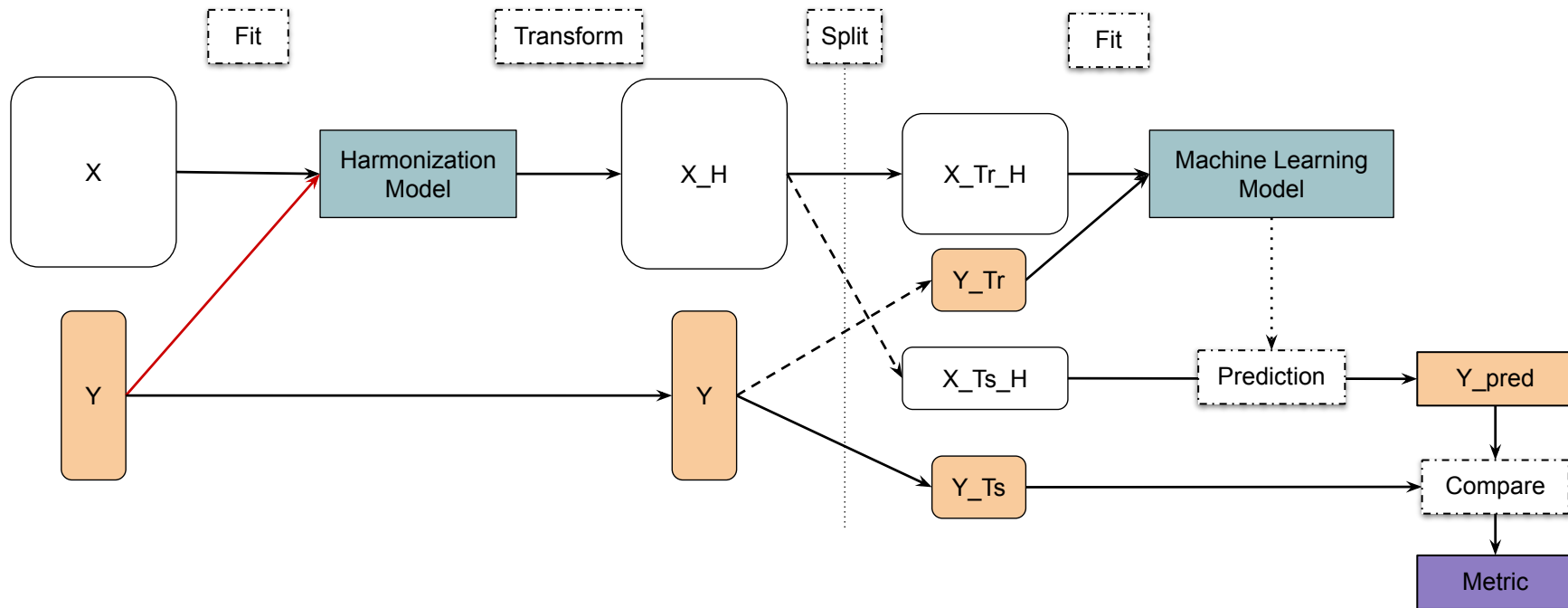
Problem characteristic:

- Real signal: No
- Effect of site: Yes
- Site-class imbalance: Yes

EoS as “signal”,
Still some performance!



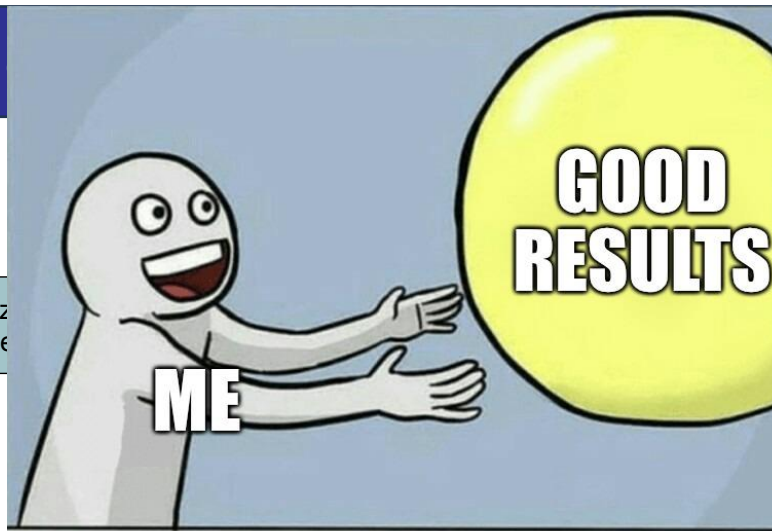
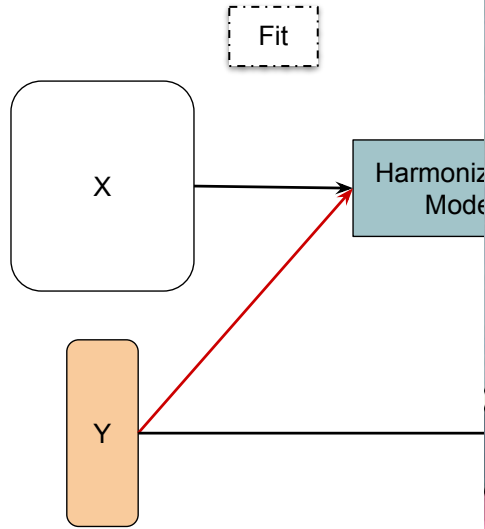
LS models: Whole data harmonization



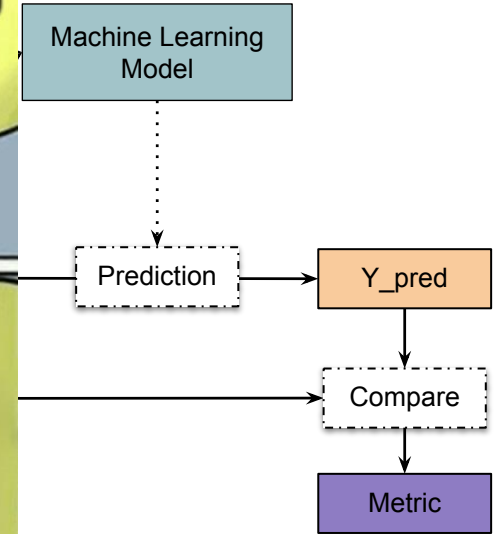
Test labels and data is used to Fit the Harmonization model. (Data leakage!)

While this is wrong in most ML applications it may be used in statistical analysis.

LS mod



nization



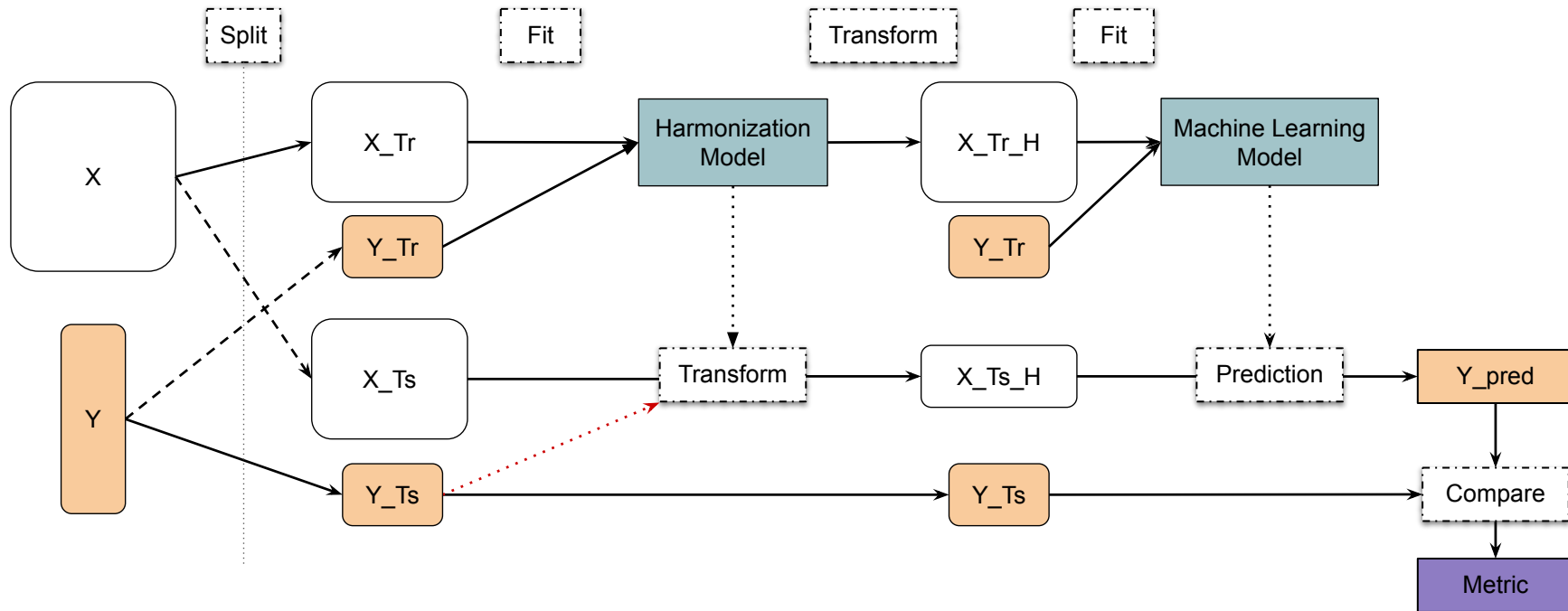
Test labels and data is used to F
While this is wrong in most ML a

sis.



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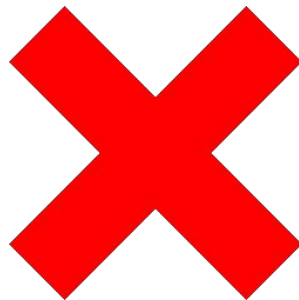
LS models: Preserving target as covariate.



If we preserve the target variance, we need to use the test labels to Transform the test data! (Test target leakage)

ML and LS models assumptions.

ML aims to predict the test target on unseen data.



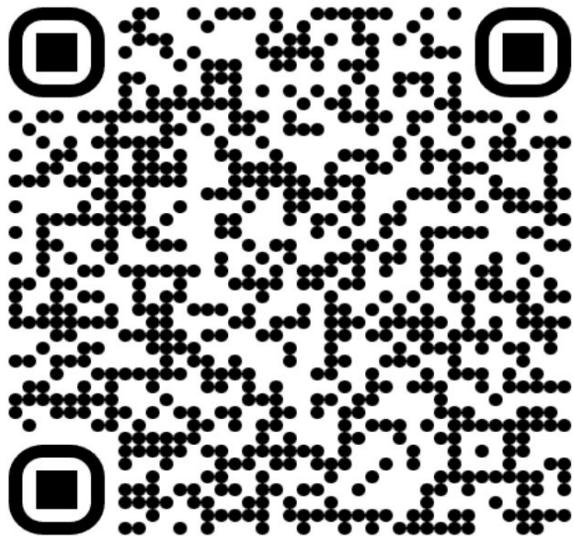
LS models need the test target to preserve the variance when is not share across sites.

What happened if we do not provide the test target to ComBat?



LS models and site-imbalance data

How will our LS model behave with site-imbalance data?

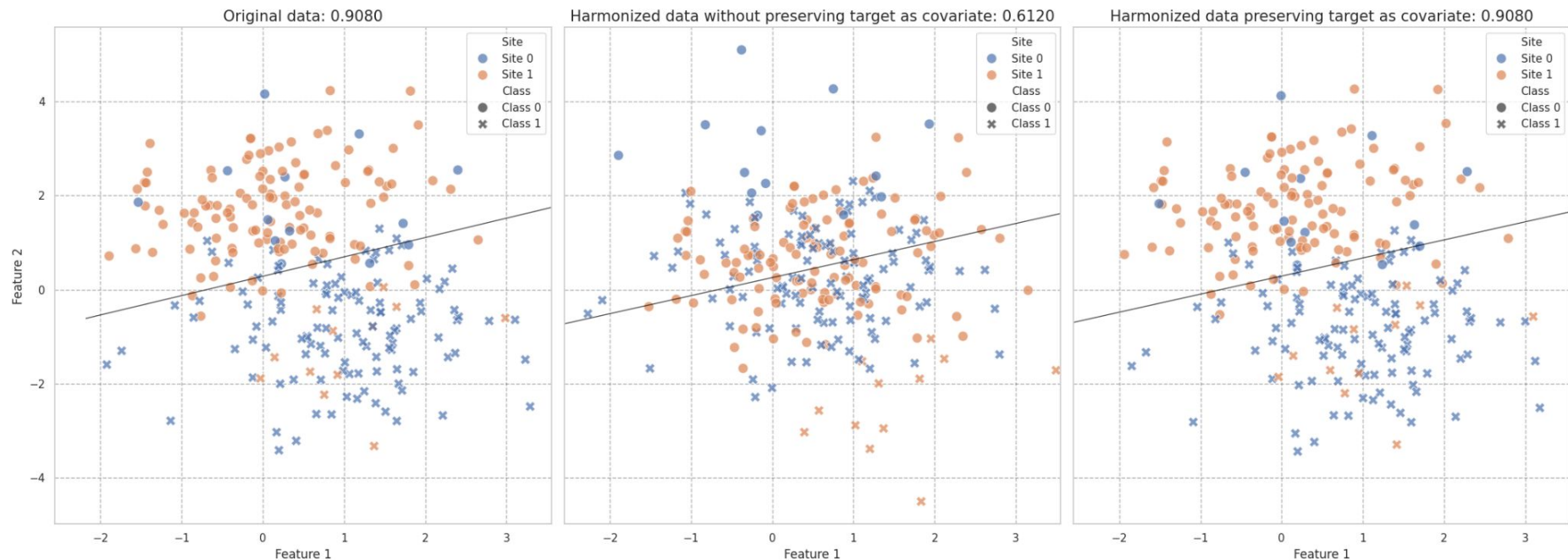


`notebooks/block03/ComBat_in_imbalance_classes.ipynb`

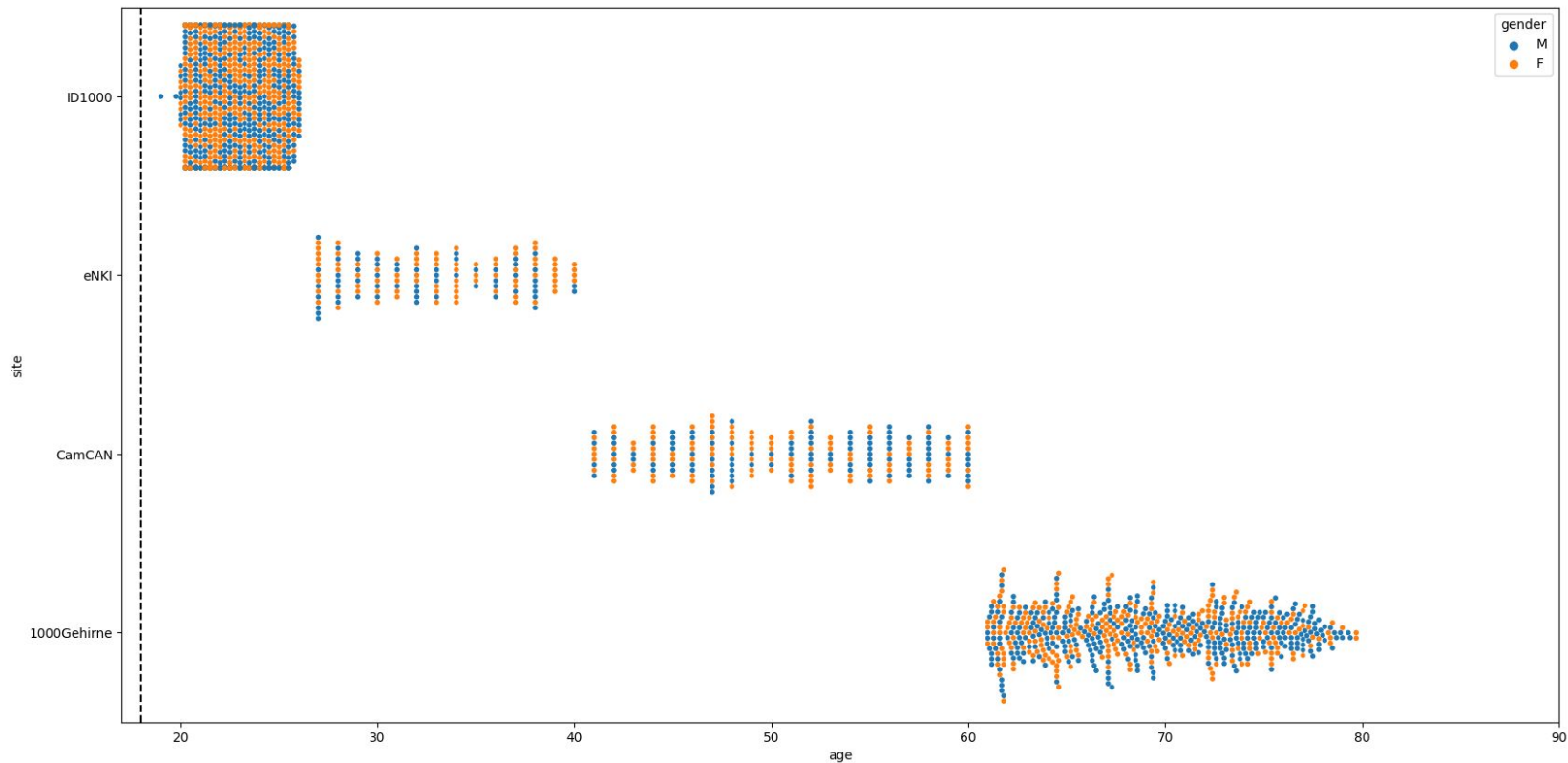


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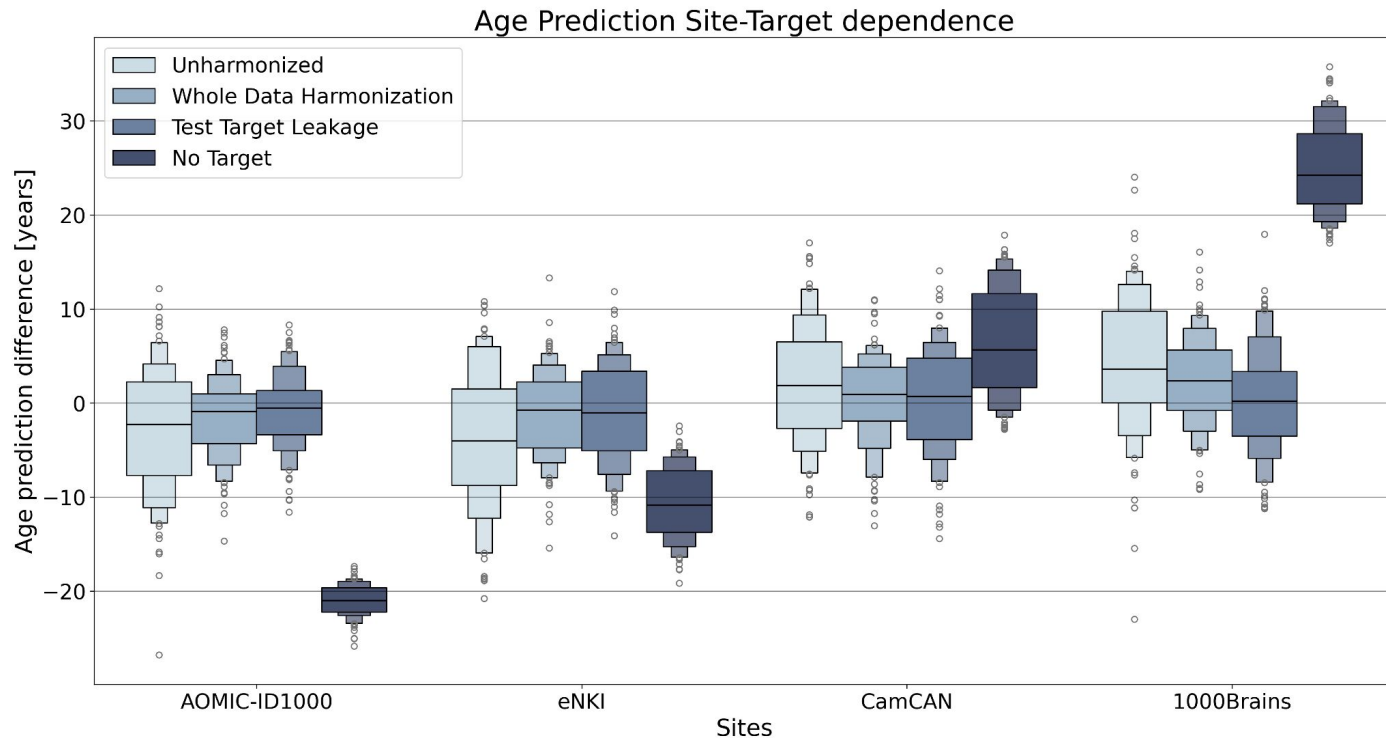
ML and ComBat without preserving target



Age regression - site target dependence



Age regression - site target dependence



Nieto, Nicolás, et al. "Impact of leakage on data harmonization in machine learning pipelines in class imbalance across sites." Neurocomputing (2026): 133146.



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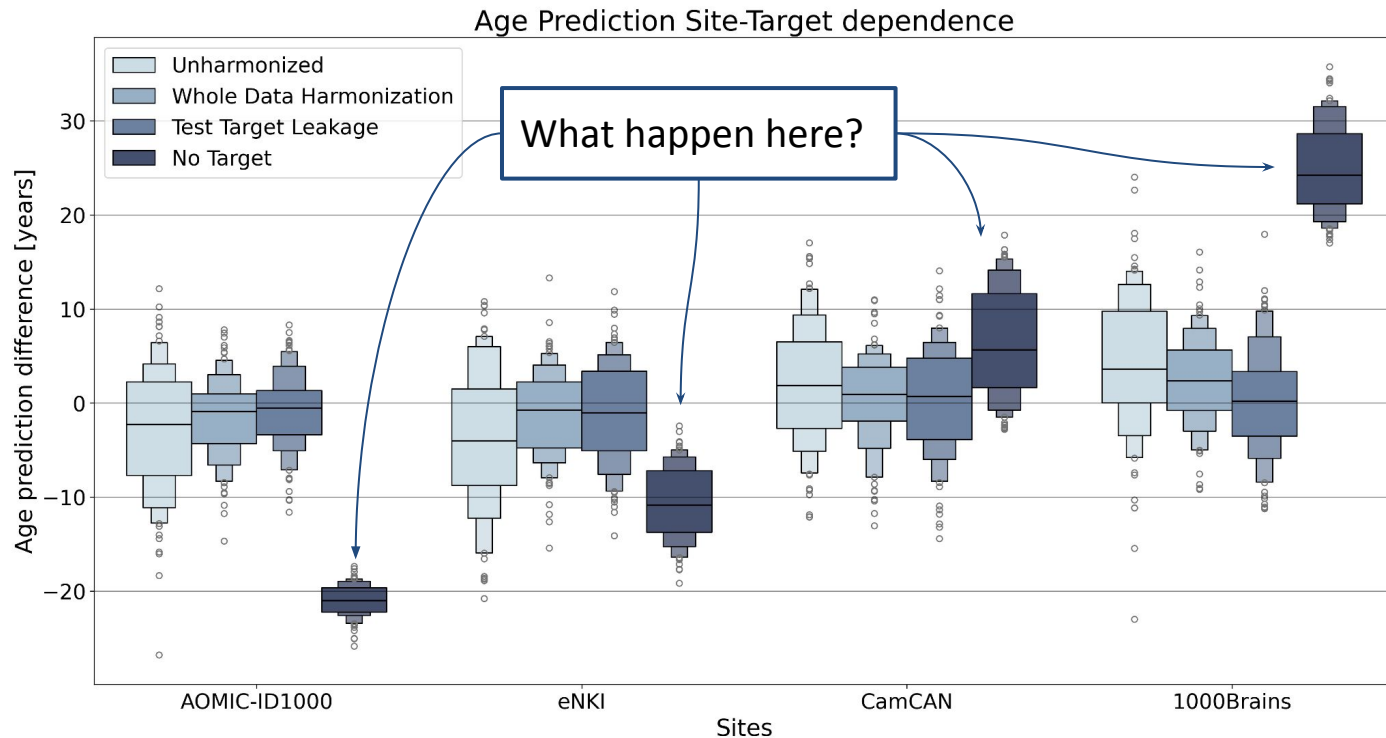
Age regression - site target dependence



Site A



Site B



Nieto, Nicolás, et al. "Impact of leakage on data harmonization in machine learning pipelines in class imbalance across sites." Neurocomputing (2026): 133146.



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Site-target dependence

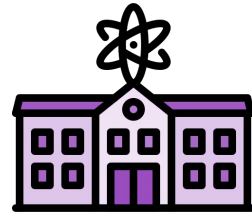
Site-target dependence



Site A



Site B



Random
noise

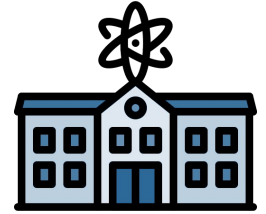
EoS

Biological
relevant
signal

EoS and signal are
confounded



Everything is removed
if not preserved



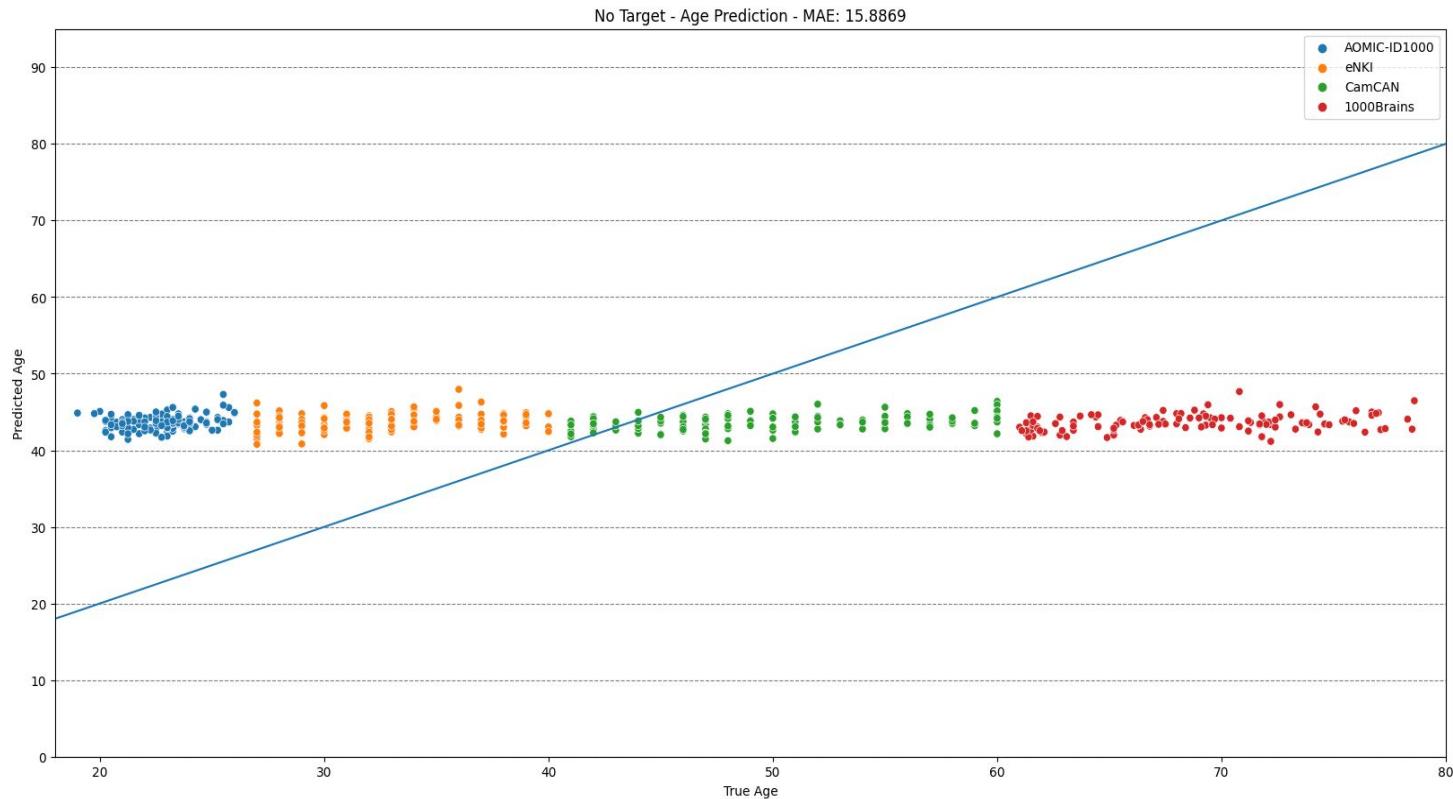
Random
noise

EoS

Biological
relevant
signal



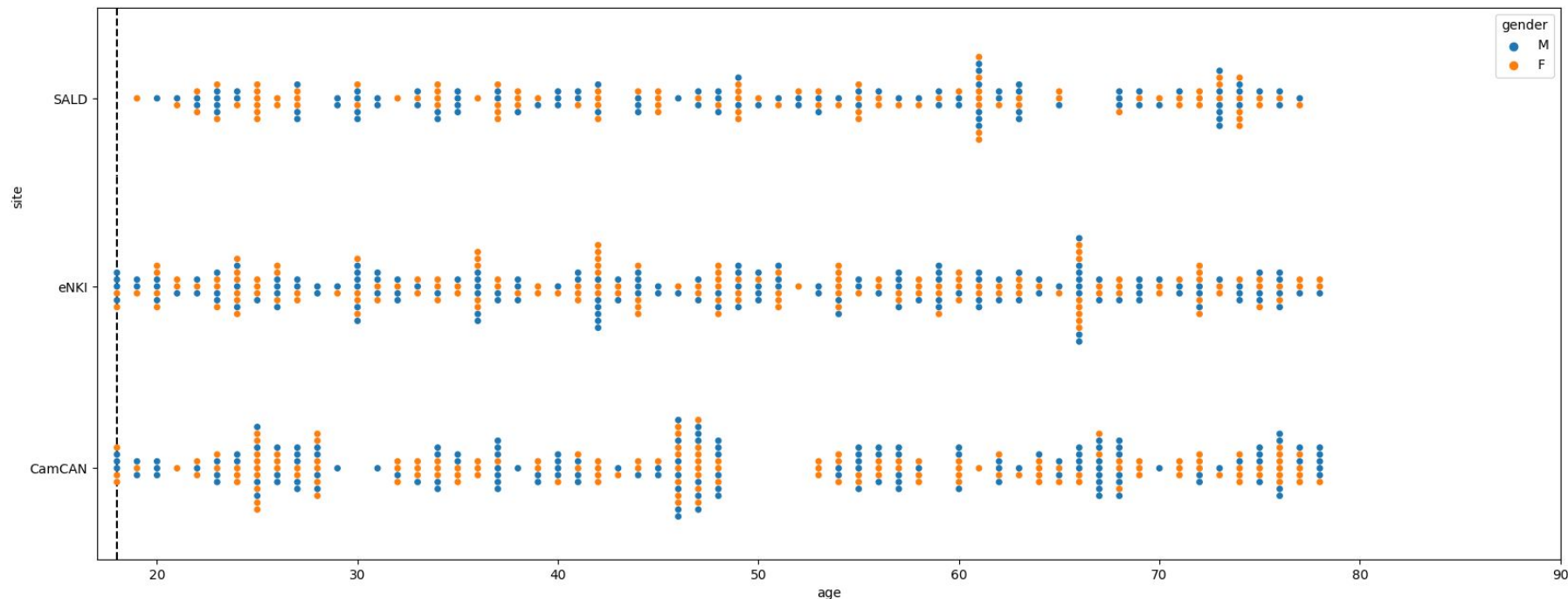
Age regression - site target dependence



Age regression - site target dependence



Age regression - site target independence



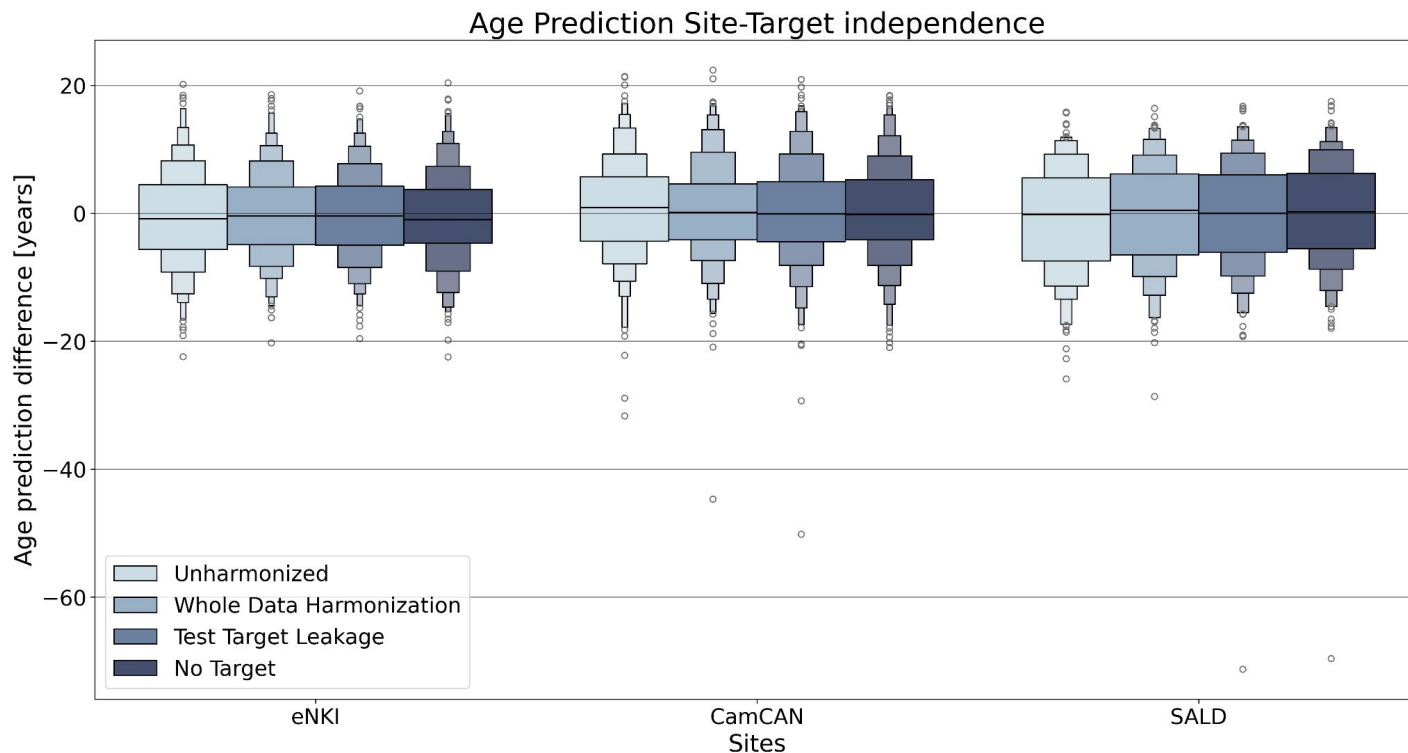
Age regression - site target independence



Site A



Site B

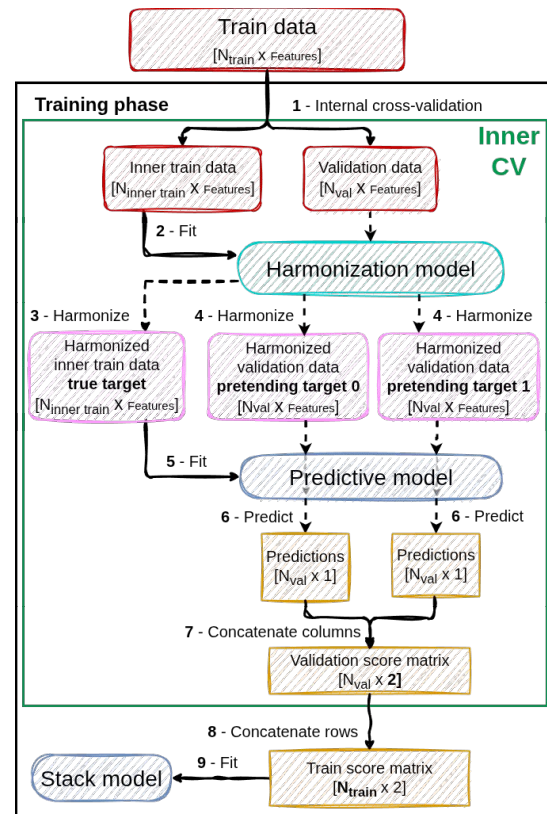


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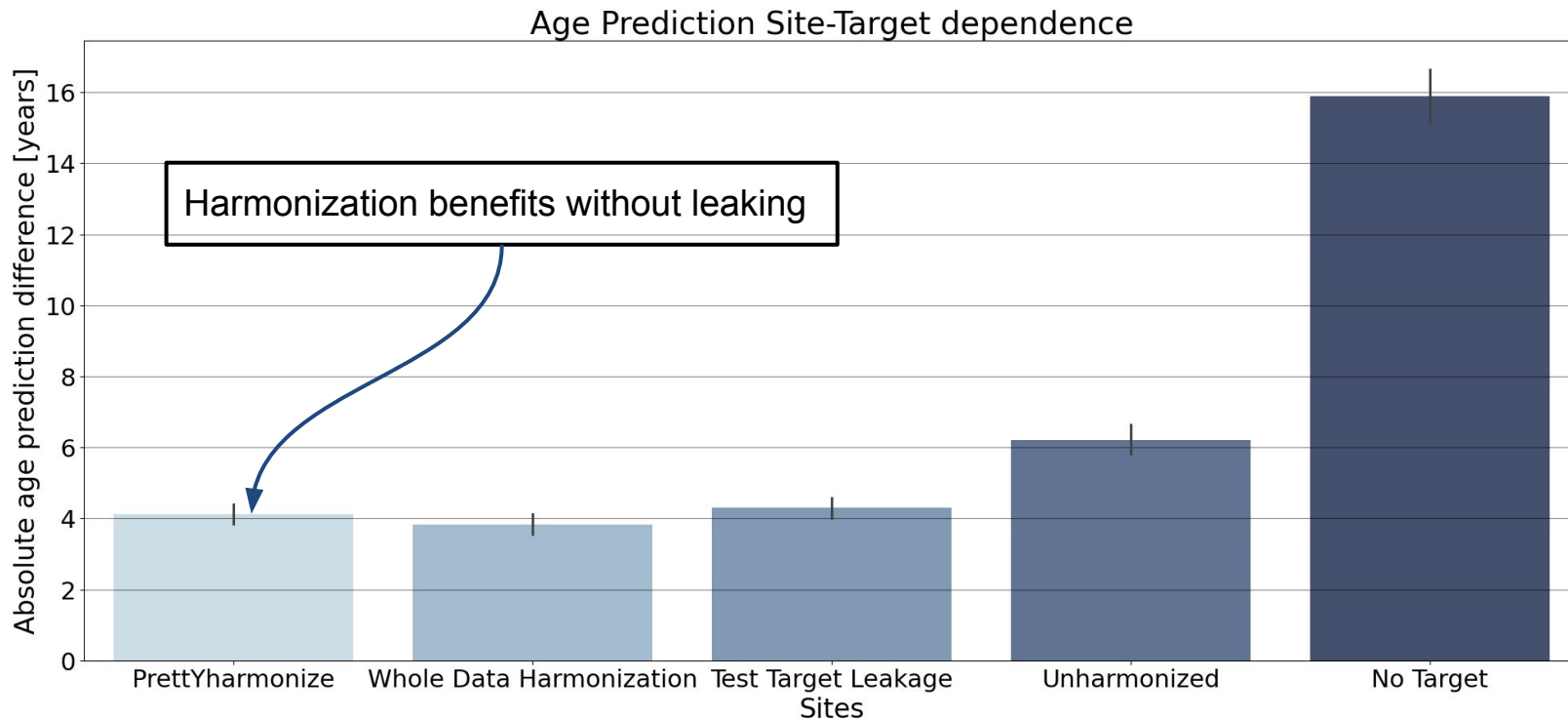
Alternative? PrettYharmonize

PrettYharmonize, to integrate ComBat-based harmonization to ML workflows preventing data leakage by design, by using a **stack ensemble** method and **pretended targets**.

Nieto, Nicolás, et al. "Impact of leakage on data harmonization in machine learning pipelines in class imbalance across sites." Neurocomputing (2026): 133146.



Alternative? PrettYharmonize



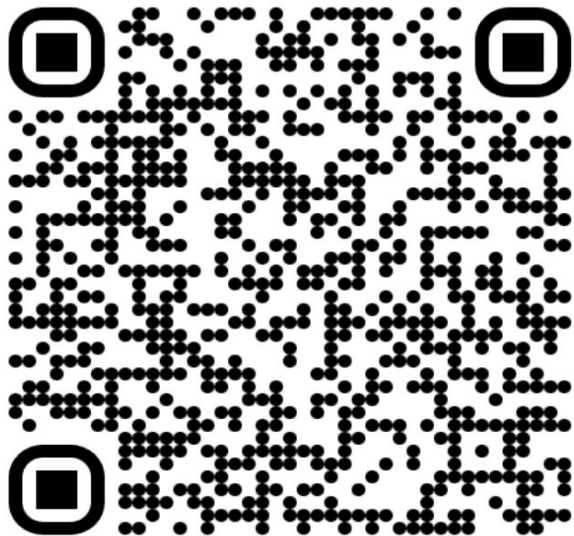
Nieto, Nicolás, et al. "Impact of leakage on data harmonization in machine learning pipelines in class imbalance across sites." Neurocomputing (2026): 133146.



ML in Multi Site data - Notebook

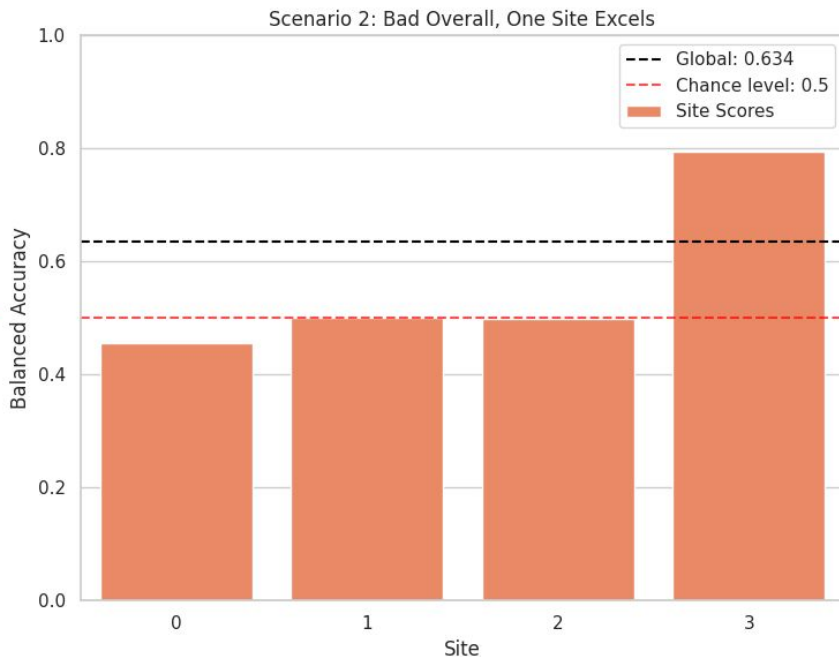
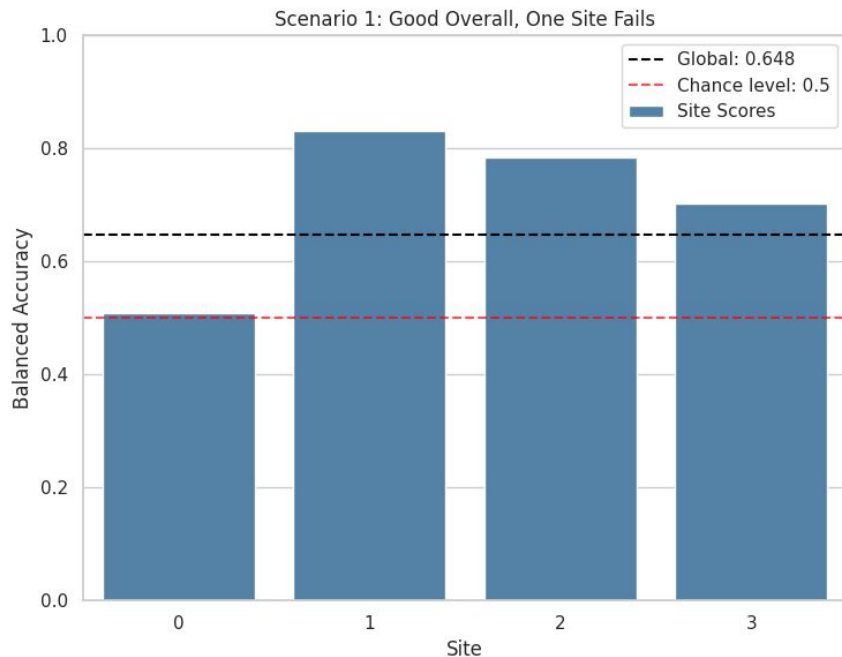
When we combine sites, is important to check if we have a systematic bias, performing poorly in some sites and very good in others.

`notebooks/block03/B3_N3_Metrics_by_Site.ipynb`



Performance metrics for multi site data.

Reporting site-wise metrics helps us detect this biases.



Takeaways

We usually think the EoS as noise, but to our ML model can act “signal”. Knowing the class imbalance across site can help us better understand our problem and what expect from the harmonization.

If the aim of our model it to predict samples (in the future), we cannot provide the target information to our data harmonization model.

Multisite data can also hide other sources of bias, like performing systematically poorly in one site and good in the other sites.



MULTISITE DATA

We usually think of
Knowing the characteristics of
our problem area

If the aim of our study is to
provide the target population
need to integrate

Multisite data collection
systematically

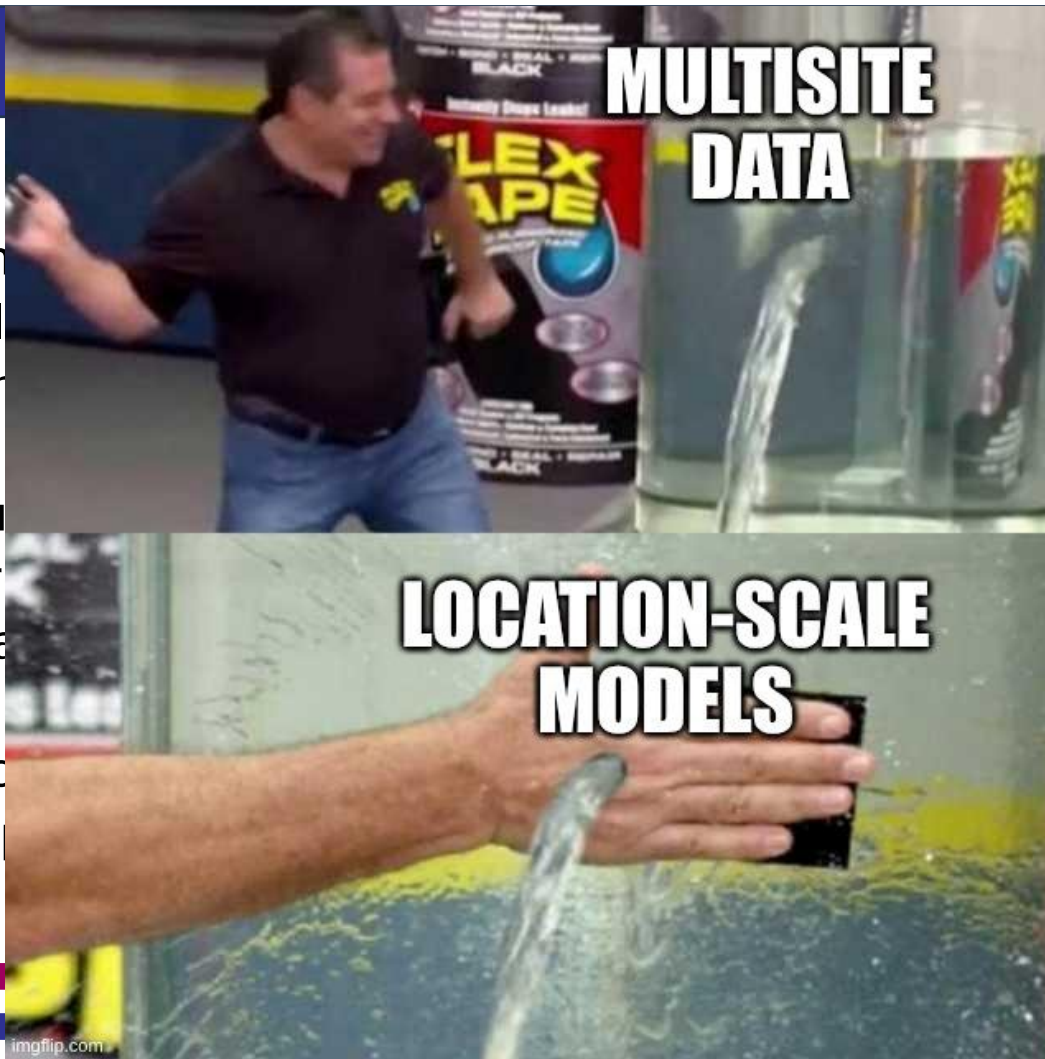
act “signal”.
understand

e cannot
del, and we

ming

LOCATION-SCALE MODELS

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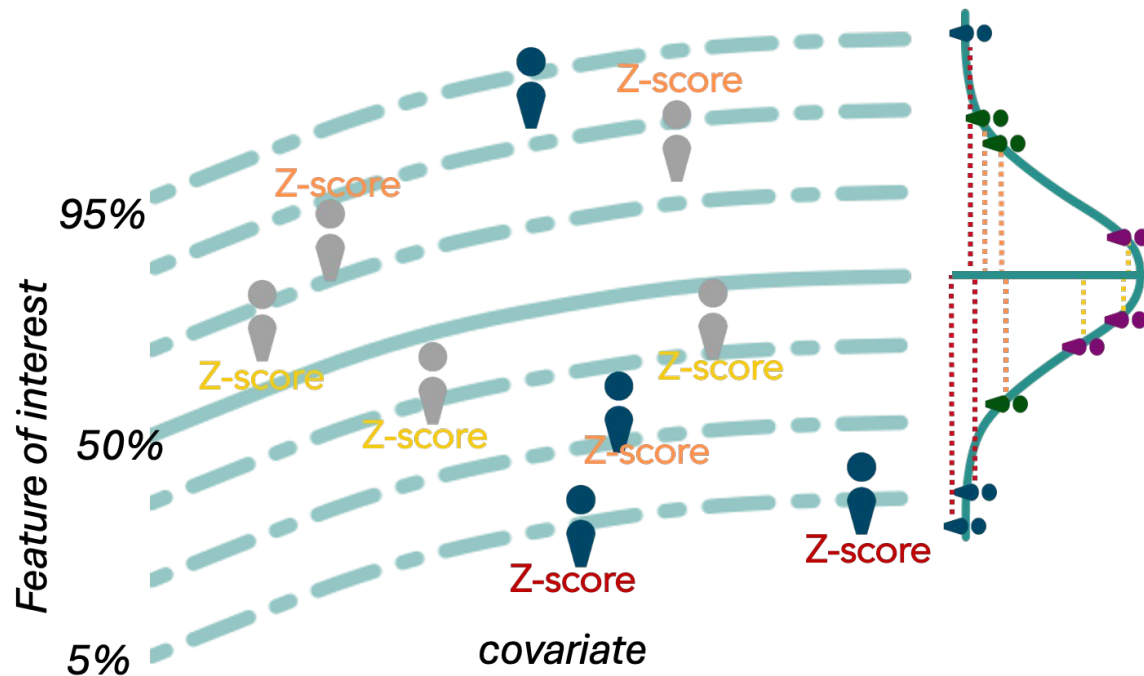
BLOCK 4



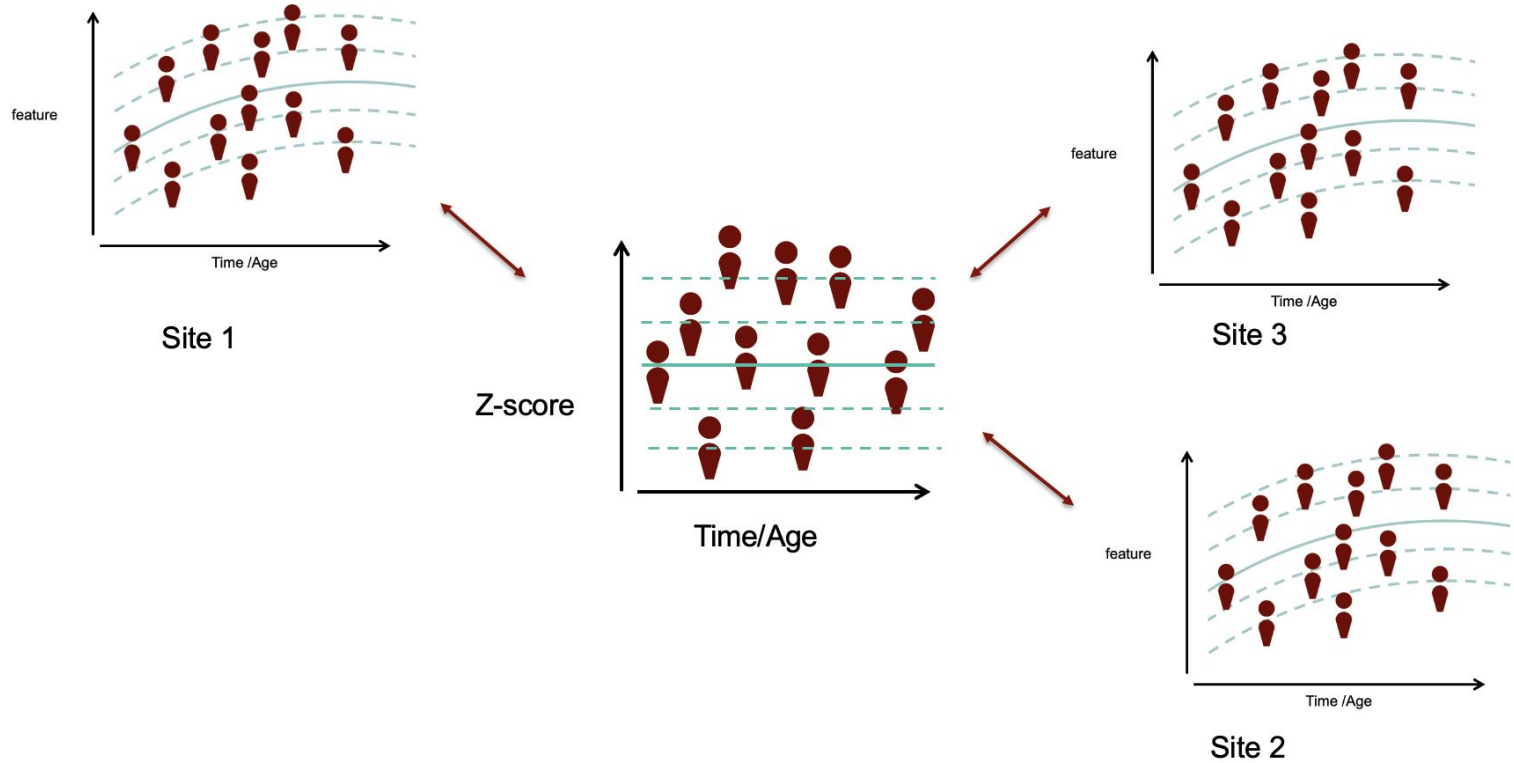
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NORMATIVE MODELLING

- Normalisation and harmonisation framework
- Variation in data is explained by covariates and expressed by centiles
- Everything that is put into a normative model is harmonized out.

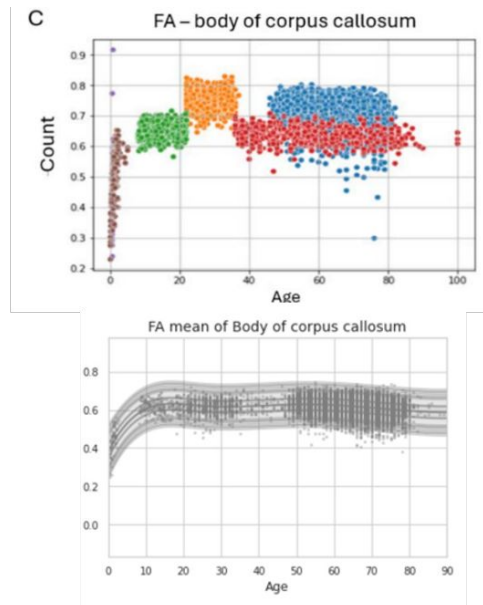
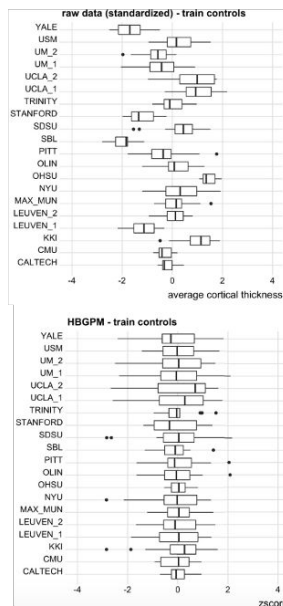


NORMATIVE MODELLING



NORMATIVE MODELLING

- Example applications



Bayer, J. M. M. *et al.* Accommodating site variation in neuroimaging data using normative and hierarchical Bayesian models. *Neuroimage* **264**, 119699 (2022).



Cirstian, R. *et al.* Lifespan normative models of white matter fractional anisotropy: Applications to Early Psychosis. *bioRxiv* 2024.12.11.627897 (2024) doi:10.1101/2024.12.11.627897.

NORMATIVE MODELLING

- Normative modelling allows:
 - Multi-site harmonisation when all data is present
 - Inverting z-scores



NORMATIVE MODELLING

- Normative modelling allows:
 - Multi-site harmonisation when all data is present
 - Inverting z-scores
- Multi-site harmonisation with new sites (pre-fitted model)



NORMATIVE MODELLING

- Notebook:
 1. Fit a normative model on the fcon1000 data set (23 sites)
 2. Convert data back into normal space, plot into select covariate space
 3. Load a pre-trained model
 4. Harmonize fcon1000 data with pre-trained model data
 5. Plot original and harmonized data



NORMATIVE MODELLING

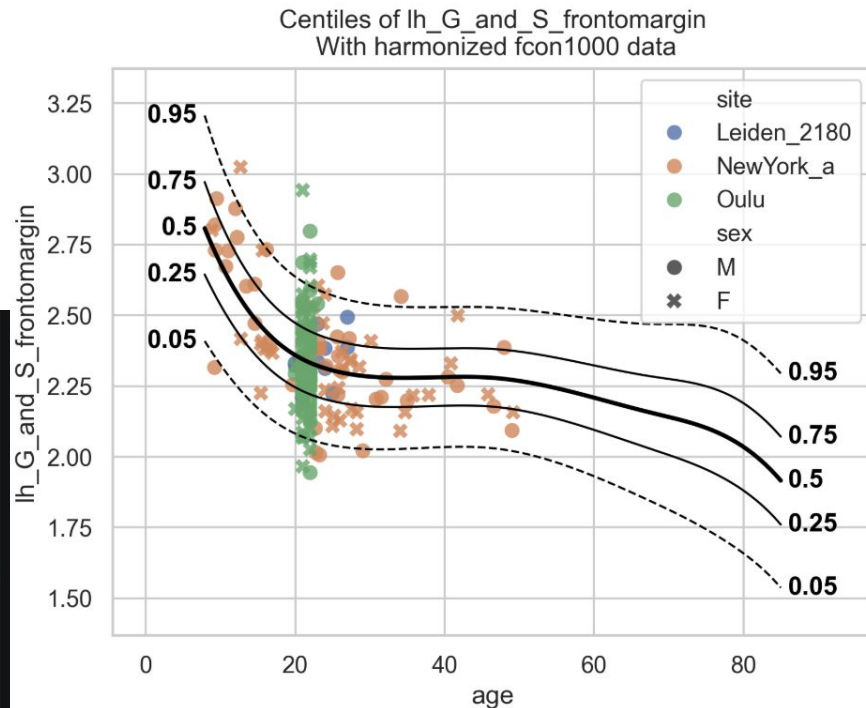
- Notebook:
 1. Fit a normative model on the fcon1000 data set (23 sites)
 2. Convert data back, plot into select covariate space

```
reference_site = "NewYork_a"
reference_sex = "F"
sites_to_plot = ["Oulu", "Leiden_2180"]

available_sites = list(model2.batch_effects_maps["site"].keys())
available_sex = list(model2.batch_effects_maps["sex"].keys())

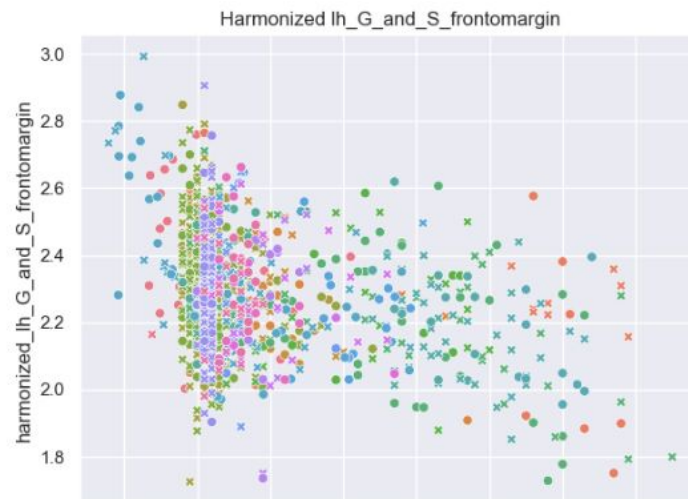
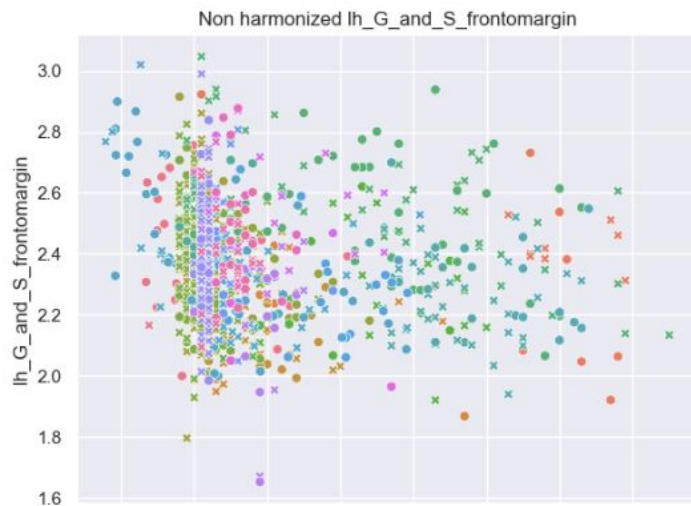
unknown_sites = [s for s in [reference_site] + sites_to_plot if s not in available_sites]
if unknown_sites:
    raise ValueError(f"Unknown site(s) for this fitted model: {unknown_sites}\nAvailable ex
if reference_sex not in available_sex:
    raise ValueError(f"Unknown sex '{reference_sex}'. Available: {available_sex}")

plot_batch_effects = {
    "site": [reference_site] + sites_to_plot,
    "sex": [reference_sex, "M" if reference_sex == "F" else "F"],
}
```



NORMATIVE MODELLING

- Notebook:
 3. Load a pre-trained model
 4. Harmonize fcon1000 data with pre-trained model data
 5. Plot original and harmonized data



Video on new harmonization methods



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IQM Harmonization - Light Version

- Batch labels used only for IQM selection, not for correction.
- IQMs chosen to capture technical variation while avoiding biological/outcome-related effects.
- Harmonisation performed using linear mixed-effects models.
- Feature-specific correction may reduce overcorrection, particularly in longitudinal and test–retest studies.
- Particularly suited to small longitudinal, repeated-measures, and test–retest datasets.

More details: Notebook B4 N1

DPUK PET-MR dataset - IQMs from T1w MRI scans

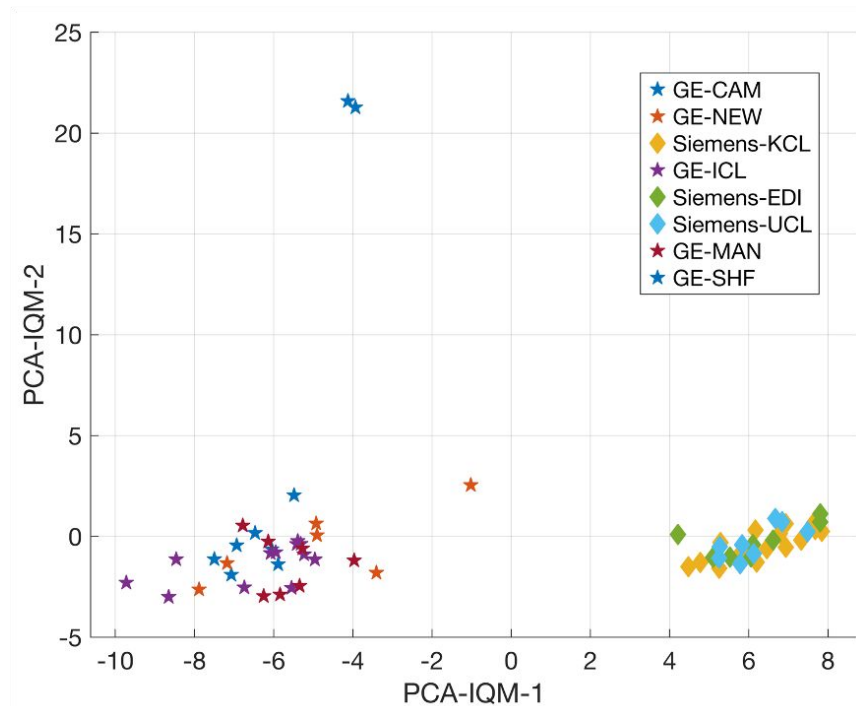


Image Harmonization - Histogram matching

- Quick and easy way to 'calibrate' one image to another
- Match intensity distribution of two images

[More details: Notebook B4 N2](#)

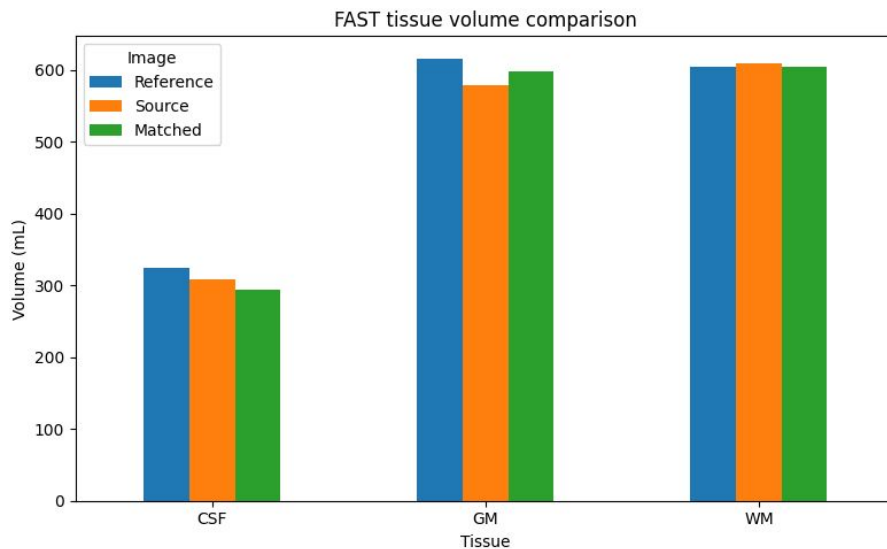
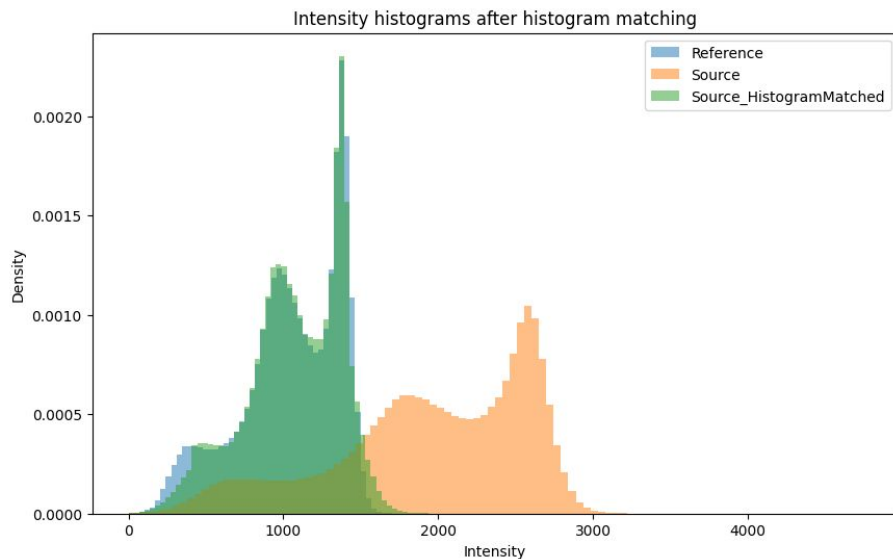
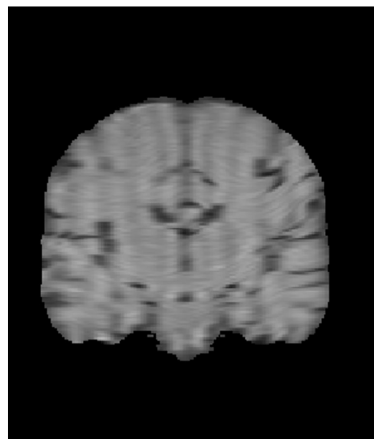


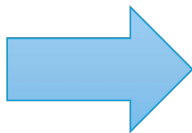
Image Harmonization - SynthSR

- Works across different contrasts, resolutions, and orientations.
- Generates synthetic high-resolution MPRAGE-like images.
- Facilitates contrast harmonisation and analysis of low-resolution clinical scans.
- Availability: <https://surfer.nmr.mgh.harvard.edu/fswiki/SynthSR>

Low-res clinical scan



Siemens 1.5T
Voxel resolution: **0.7x0.7x6**



SynthSR applied



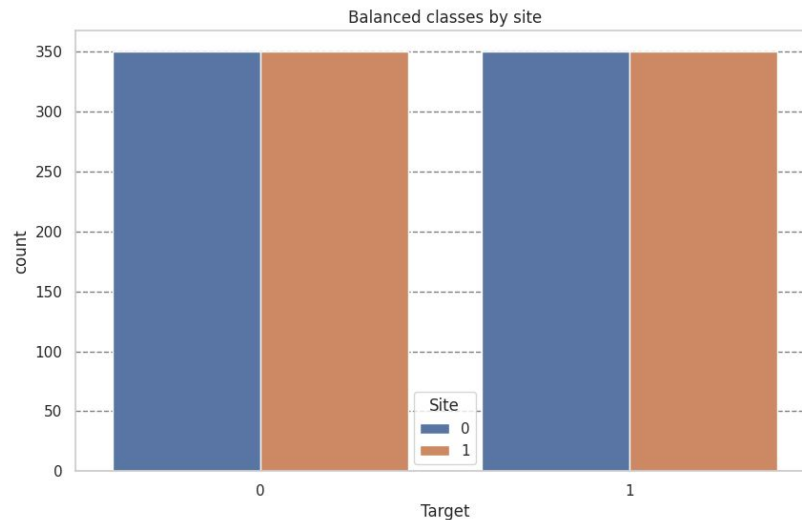
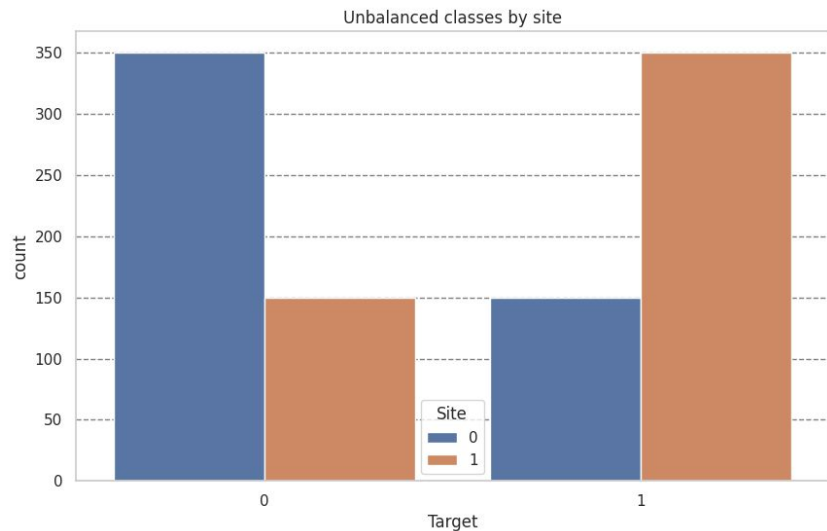
Synthetic version
Voxel resolution: **1x1x1**

**For list of current advanced
image harmonisation methods
see Notebook B4 N2**



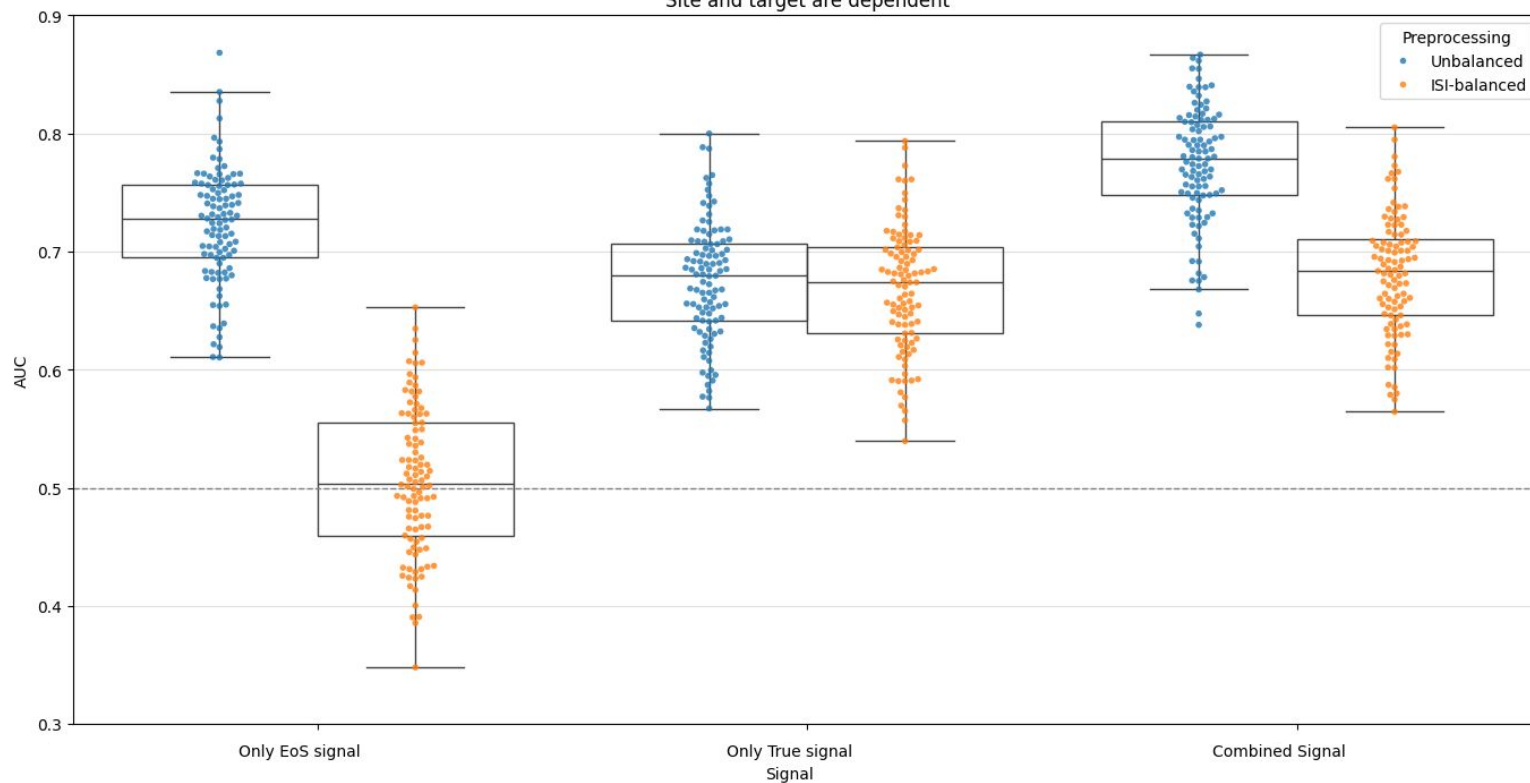
Interpolation methods

What if we remove the site-target dependence and our ML model do not use the EoS to “boost” its prediction? How? Balancing the sites!



Interpolation methods

Model AUC under different signals and preprocessing strategies
Site and target are dependent



Time for Questions!

Johanna Bayer

Johanna.Bayer@radboudumc.nl

Jacob Turnbull

jacob.turnbull@ndcn.ox.ac.uk

Emma Prevot

prevot@stats.ox.ac.uk

Nicolás Nieto

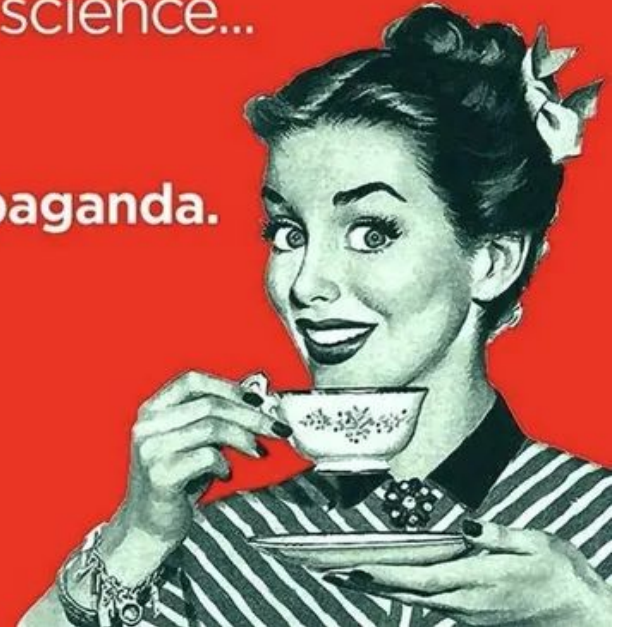
n.nieto@fz-juelich.de

Gaurav Bhalerao

gaurav.bhalerao@ndcn.ox.ac.uk

If you cannot question it,
it's not science...

it's
propaganda.



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References

Adjusting batch effects in microarray expression data using empirical Bayes methods

W. EVAN JOHNSON, CHENG LI*

*Department of Biostatistics and Computational Biology,
Dana-Farber Cancer Institute, Boston, MA, USA and Department of Biostatistics,
Harvard School of Public Health, Boston, MA, USA*
cli@hsph.harvard.edu



Bayer, J. M. M. *et al.* Accommodating site variation in neuroimaging data using normative and hierarchical Bayesian models. *Neuroimage* **264**, 119699 (2022).

Harmonization of multi-site diffusion tensor imaging data

Jean-Philippe Fortin^{1,*}, Drew Parker², Birkan Tunç², Takanori Watanabe², Mark A. Jernigan³, Kosha Ruparel⁴, David R. Roalf⁴, Theodore D. Satterthwaite⁴, Ruben C. Gur^{3,4}, Raquel E. Gur^{3,4}, Robert T. Schultz⁵, Ragini Verma^{2,†}, and Russell T. Shinohara^{1,†}

¹Department of Biostatistics and Epidemiology, Perelman School of Medicine, University of Pennsylvania
²Center for Biomedical Image Computing and Analytics, Department of Radiology, Perelman School of Medicine, University of Pennsylvania

³Department of Radiology, Perelman School of Medicine, University of Pennsylvania

⁴Department of Psychiatry, Perelman School of Medicine, University of Pennsylvania

⁵Center for Autism Research, The Children's Hospital of Philadelphia



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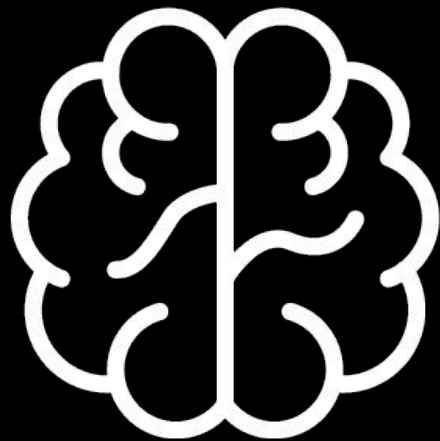


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